

Accepted Service Adaptation: Exact Parametric Quotients and Minimal Additional Writable Memory

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Abstract. Renewal agreements can reach the same public state with different accepted continuation obligations. We study exact representation under one additive commitment and separable concave rewards. Probability weighting yields classical laminar allocation, and normalized memoization gives the public-graph recursion. For quadratic rewards, the global knot universe is linear and explicit vertex tables have tight quadratic size. Exact closure extends to continuous piecewise-quadratic rewards, including marginal jumps, with rational complexity bounds and shared storage. Classical behavioral minimization identifies the minimum additional writable alphabet conditional on free public observation; combined states and read-only storage are counted separately. A heterogeneous renewal model yields deterministic and globally optimized randomized memory frontiers. All but the highest randomized level can be anchored at branch caps; the remaining continuous level is optimized exactly. Symbolic staircase completion permits linear work per budget layer for both complete frontiers without discretizing service. An exact example establishes that cap-only codebooks are suboptimal. Randomization improves expected participation but cannot improve any budget under pathwise participation in this renewal architecture. Exact controller replay, exhaustive comparisons, complete parametric-path tests, and same-input computational comparisons support the analysis. Earlier switching, shared-table, and deployment results are preserved.

Keywords: continuation participation; contractual memory; parametric resource allocation.

Subject classifications: Programming: stochastic resource allocation; Dynamic programming/optimal control: contractual memory.

Area of review: Stochastic Models.

1 Introduction

A service agreement can return to the same public operating condition after different renewal reviews. Those histories need not permit the same future service: a customer who could have exited at an earlier review may have accepted a different continuation obligation. A provider that retains only the current public condition can therefore lose value even with Markovian demand, fixed prices, and no private information. The operational questions are how to represent the complete accepted continuation response, how much history an optimal protocol must retain, and whether randomization can compensate for a restricted record.

We study positive additive payments, separable strictly concave operating rewards, interval controls, and an exogenous recombining public graph. The single-quadratic case exposes the structure; a further theorem permits changing curvature and downward jumps in marginal reward through continuous piecewise-quadratic functions. There is one scalar commitment, and the exact compiler excludes intertemporal switching and coupled vector promises. Participation caps reflect a continuation price at public-vertex-specific barriers. A backward construction computes the complete parameterized response before the root promise is selected, without first obtaining an extensive-form optimum.

The optimization lineage is explicit. Probability weighting converts the unfolding into classical separable laminar allocation, as in Mjelde (1983) and Tang (1990). Normalized memoization yields the same public-graph recurrence. Parametric piecewise-quadratic flow and polymatroid sensitivity are established subjects (Klimm and Warode, 2021; Harks et al., 2018). We give an explicit convex-cost-flow representation of the unfolding and distinguish its primal supply parameter from the dual price used by the compiler. Neither scalar marginal allocation, generic parametricity, nor memoization is claimed as a new principle. The contribution is a quantitative event budget measured in the compact public input, together with exact rational preprocessing and representation choices for that complete response.

The quadratic specialization has a linear global knot universe and a matching quadratic number of pieces across separately materialized vertex tables. A chain attains both orders. These are explicit-output bounds, not lower bounds for every possible shared representation. A shared arithmetic circuit retains local responses and graph links instead of all vertex tables, using linear space in the piecewise input after compilation at the cost of slower queries. The piecewise-quadratic extension bounds events by the local piece count plus the number of public caps. It covers genuine marginal jumps, retains exact reflection and rational certificates, and does not enlarge the alphabet of carried cap prices. Thus preprocessing space, persistent read-only representation, and execution memory are distinct resources.

Execution is governed by a finite deterministic output transducer driven by public transitions. Backward state minimization is classical (Bean et al., 1987; Givan et al., 2003; Peyrière, 2023). The contract-specific conclusions are finite reachability from continuous controls, ordered behavioral classes, and a worst-case linear additional writable alphabet under a stated public-observation interface. We report that alphabet separately from combined public-state/class pairs and read-only coefficients. Distinct numerical prices can produce identical future behavior and therefore share a symbol. Conversely, a recombining renewal family requires a distinct symbol for each distinct optimal terminal tier. Finite policy graphs already represent dynamic contracts (Zhang, 2012); here the graph is exact for a fixed promise in a finite-horizon public-information model, not an approximation for adverse selection.

The renewal family provides both a deterministic memory-value frontier and a randomized extension. The deterministic optimum has contiguous cap-ordered cells under heterogeneous probabilities and costs. For a fixed randomized terminal codebook, adjacent lotteries give an exact concave-envelope formula. An exactly solved three-branch example shows that two symbols can have strictly smaller loss with randomization, even though three symbols remain necessary for the exact optimum. This result applies when participation is evaluated in conditional expectation before the private draw. The corresponding lottery is not asserted to satisfy every private realization’s cap. The retained comparative static concerns ordered mean-preserving radial spreads, rather than arbitrary convex-order comparisons.

The randomized design problem can also be solved globally for quadratic terminal rewards and heterogeneous quadratic intermediate costs. We prove that an optimal codebook has cap-anchored levels except for its highest level, which may be genuinely continuous. An ordered dynamic program with an analytically optimized terminal interval then computes the complete restricted-budget frontier in polynomial exact arithmetic. A heterogeneous example demonstrates that constraining all levels to branch caps misses the optimum. Under participation imposed after every private draw, a separate theorem shows that randomization cannot improve the deterministic frontier. These results turn the distinction between expected and realized participation into a directly computable service-design choice.

The complete frontier admits a further structural acceleration. Its interpolation costs and terminal costs have ordered minimizers, even after the highest service level is optimized continuously. This permits both participation frontiers to be computed by monotone search rather than quadratic scans. A symbolic completion also preserves ordered minima in every submatrix, enabling an implemented classical linear-search algorithm for each budget layer. The search principle is classical; the new argument establishes its hypotheses for heterogeneous contractual costs and for the minimized terminal array. An exact same-input comparison isolates this improvement, preserves off-cap optimal service levels, and measures oracle work separately from machine-dependent runtime.

Computation checks the claimed parametric object rather than relying on an expensive unfolding as a novelty argument. The retained laminar, convex, public-promise, and repeated-query studies remain available. New exact tests cover changing curvature, marginal jumps, shared-circuit evaluation, root inversion, and bounded-conjugate certificates. An independently coded exhaustive active-set method constructs whole parametric paths on small unfolded quadratic problems and compares every affine region. This reference is deliberately small and is not a timing benchmark of a published specialized solver. The results provide reproducible mathematical and implementation evidence, not calibrated service data or a claim of universal computational superiority.

The broader accepted-adaptation theory on switching, comparator-adjusted rents, continuous states, and certified deployment is retained with its original assumptions and evidence. The present article adds exact closure, a storage–query tradeoff, and a contractual role for restricted-memory lotteries while preserving the earlier quadratic, deterministic, and shared-table results as identifiable parts of the theory.

2 Service Agreements and Their Laminar Allocation Form

2.1 Public primitives and accepted continuations

Let $G = (\mathcal{V}, \mathcal{E})$ be a finite, rooted, acyclic public graph. Vertices include date; all are reachable. Write $N = |\mathcal{V}|$, $M = |\mathcal{E}|$, and let $p_{vw} > 0$ be the exogenous probability on edge (v, w) , with probabilities summing to one at a nonterminal vertex. The discount factor is $0 < \beta \leq 1$. An occurrence h is a root-to-vertex history in the finite unfolding $\mathcal{T}$; its public vertex is $v(h)$ and its discounted reach weight is $w_h = \beta^{t(h)} \Pr(h) > 0$. Write $j \succeq h$ for a descendant, including h , and $H = |\mathcal{T}|$.

At occurrence h of v , the provider chooses a tier $x_h \in [l_v, h_v] \subseteq [0, 1]$. The payment coefficient $a_v > 0$ is fixed, and the operating reward is $g_v(x) = r_v x - q_v x^2/2$, with $q_v > 0$. Customer utility is a publicly specified service-benefit term minus $a_v x$. For example, fixed stock s_z yields

$$u(z, s_z, x) = v \mathbb{E} \min(s_z, D_z) - a_z x, \quad a_z = r_z^{\text{premium}} - \kappa \mathbb{E}(D_z - s_z)^+ > 0. \quad (1)$$

Stock may be stipulated, or identified by the service and cost commitments in the retained stock-identification results. The customer can leave before each modeled review; the provider commits to its contingent protocol. Retaining the customer requires the conditional service benefit minus remaining payment to exceed the public outside continuation. Assume their difference is a vertex-sufficient cap B_v . The accepted optimization is

$$\begin{aligned} J(b_0) = \max_x \quad & \sum_{h \in \mathcal{T}} w_h g_{v(h)}(x_h) \\ \text{subject to} \quad & Y_h := \sum_{j \succeq h} \frac{w_j}{w_h} a_{v(j)} x_j \leq B_{v(h)} \quad (h \in \mathcal{T}), \quad Y_0 = b_0, \\ & l_{v(h)} \leq x_h \leq h_{v(h)}. \end{aligned} \quad (2)$$

The root has a cap as well as the equality promise. No privately informed reports, endogenous termination choice, or policy-dependent transition probabilities are part of this institution. The no-switching restriction is essential to the scalar-price result; the retained general model continues to allow switching and is not replaced by (2).

The feasible remaining-payment interval is computed without unfolding:

$$L_v = a_v l_v + \beta \sum_w p_{vw} L_w, \quad U_v = \min \left\{ B_v, a_v h_v + \beta \sum_w p_{vw} U_w \right\}. \quad (3)$$

A subtree is feasible precisely when its successors are feasible and $L_v \leq U_v$. Throughout, $b_0 \in [L_0, U_0] \subseteq (-\infty, B_0]$. Convexity fills every point between the two extreme payments. Compactness gives an optimum and strict concavity gives a unique occurrence-indexed action vector.

2.2 An exact mapping, not an analogy

Proposition 2.1 (Laminar allocation equivalence). *Define $z_h = w_h a_{v(h)} x_h$. Problem (2) is exactly the separable concave allocation problem*

$$\begin{aligned} \max_z \quad & \sum_h \left[\frac{r_{v(h)}}{a_{v(h)}} z_h - \frac{q_{v(h)}}{2w_h a_{v(h)}^2} z_h^2 \right], \\ \sum_{j \succeq h} z_j \leq & w_h B_{v(h)}, \quad \sum_j z_j = b_0, \\ w_h a_{v(h)} l_{v(h)} \leq & z_h \leq w_h a_{v(h)} h_{v(h)}. \end{aligned} \tag{4}$$

The constrained descendant sets are laminar. The change of variables is invertible and preserves objective values and optimal policies.

Proof. Substitution gives each term and row of (4). Two rooted descendant sets in a tree are either disjoint or one contains the other. Positivity of $w_h a_{v(h)}$ makes the coordinate map bijective and preserves every inequality direction. $\square$

Thus the expanded primal lies in the tree-constrained allocation lineage of Mjelde (1983) and Tang (1990). Tang reports a reduction to one-variable convex subproblems; that result is part of the inherited optimization structure, not a competitor we relabel. The chain-nested allocation results of Vidal et al. (2016); Schoot Uiterkamp et al. (2021); Wu et al. (2021) concern related but different input structures. A complexity bound for an ascending chain cannot simply be assigned to arbitrary branching descendant sets.

After subtracting lower allocations, the packing region is a laminar polymatroid, and the exact total selects a base of its truncation (Electronic Companion, Section EC.1). This connects the model to Federgruen and Groenevelt (1986) and Groenevelt (1991). Harks et al. (2018) further show that sensitivity analysis for separable optimization over integral polymatroid bases is a substantial literature in its own right. Their decision set is integral and their perturbations concern costs and polymatroid parameters; our theorem concerns a continuous quadratic accepted-service model and a complete response in one dual price. We cite that work as sensitivity lineage, not as a result to be rediscovered.

2.3 What remains after memoization and generic parametric optimization

An occurrence-indexed marginal reduction may represent subtree payment in the unnormalized units z . Its entire response is the occurrence weight times a common conditional response, proved below. Dividing by that weight makes occurrences of the same public vertex identical. Memoizing those normalized responses therefore gives our graph recursion. There is no additional stochastic obstruction or different first-order condition. We do not assert that scalar marginal allocation, subtree reduction, or memoization originated here.

The closest modern parametric comparison is Klimm and Warode (2021), who construct output-polynomial algorithms for separable, continuous, piecewise-quadratic parametric minimum-cost flow. Their parameter scales network demands and their feasible set is a flow polyhedron; our parameter is the price

conjugate to an accepted-service payment and the unfolded feasible set is laminar allocation. A laminar problem may itself be reformulated before a parametric method is applied, so we make no priority claim for generic piecewise-affine solution paths or for output-sensitive parametric optimization. The contribution claimed here is narrower and quantitative: after quotienting repeated occurrences to the supplied public graph, all response changes are generated by local clipping events and at most one cap barrier per public vertex. This gives an at-most- $3N$ global knot universe, a matching $\Omega(N)$ family, a matching $\Omega(N^2)$ total-response family, and an explicit rational bit bound measured in the compact graph input. These conclusions are proved directly in Section 3.

This positioning answers five distinct questions. First, the complete response is a special structured parametric convex solution map, so pathwise parametricity itself is not claimed. Second, an unfolded laminar algorithm can compute the same path, and normalized memoization yields the same recurrence. Third, the compact quotient makes the response-size bound depend on public vertices rather than occurrence histories. Fourth, the global knot and total-storage orders are shown below to be tight for polynomial-size rational instances. Fifth, the bit bound is for the explicit coefficient-event constructor and includes exact sorting and comparison cost; it is not a generic encoding theorem for every parametric solver.

2.4 Behavioral minimization and policy-graph lineage

The reached execution object is also connected to classical minimization. Bean et al. (1987) study aggregation in dynamic programming; Givan et al. (2003) develop state equivalence and model minimization for MDPs; and Peyrière (2023) gives a minimum-state equivalent Moore-machine construction. Our backward equality of current outputs and successor classes is an instance of that classical deterministic behavioral-minimization principle. Section 5 therefore states the reached price system as an output transducer first, applies standard minimization, and then isolates the contract-specific results: finite reachability from continuous controls, ordered contiguous behavior classes, separate writable/read-only accounting, and a linear contractual lower-bound family.

Promised continuation utility is a standard recursive state in dynamic contracting (Spear and Srivastava, 1987; Thomas and Worrall, 1988; Chen et al., 2020). Zhang (2012) is especially relevant because it uses finite policy graphs to represent continuation contracts and payoff frontiers in an infinite-horizon adverse-selection model with Markov information. That policy graph is an approximation device for a different information and incentive problem. Our public-information model is finite horizon, has no private reports, and compiles an exact transducer for a fixed root promise. The comparison is nevertheless conceptually close: the present result can be read as an exact policy-graph compression theorem for this scalar public-information subclass.

Policy-graph decomposition and stochastic decomposition already avoid naive scenario enumeration (Pereira and Pinto, 1991; Girardeau et al., 2015; Dowson, 2020). We compare an explicit primal promise-grid method on the public graph rather than treat the mere use of a graph as a contribution. Our exact closure depends on one scalar additive commitment and separable quadratics; general convex policy-graph decomposition has a broader domain. The supporting cuts for fixed tables are also instances of standard dual decomposition (Benders, 1962; Boyd and Vandenberghe, 2004), not a new general finite-convergence theorem.

Input-size convention. An explicit-tree representation has H local records and must at least expose those records to an occurrence-indexed implementation. Our compact representation has N vertices and M edges. This is an input/output comparison, not a lower bound against every algorithm that accepts a compact graph. The independent tree code is retained as an exact cross-check, not as a stand-in for the strongest classical or modern specialized solver.

3 A Tight Quantitative Quotient of the History Tree

3.1 Conditional responses and cap reflection

For an occurrence h of v , let $H_h(\eta)$ be the maximum conditional reward minus ηY_h , imposing all subtree caps but not an exact payment equality at h . Let $H_h^{\text{pre}}(\eta)$ remove only the cap at h , retaining every successor cap. Conditional rewards use weights w_j/w_h . Define $R_h(\eta)$ as the payment of the unique maximizer of $H_h(\eta)$; use $d_h(\eta)$ for the pre-cap payment. These definitions separate a removed current cap from the constrained continuation problems below it.

Theorem 3.1 (Occurrence-to-vertex quotient and finite response closure). *Under the assumptions of Section 2, occurrences h, h' of a public vertex v with the same incoming price η have identical normalized optimal continuation plans. In particular, $H_h = H_{h'} =: H_v$, $H_h^{\text{pre}} = H_{h'}^{\text{pre}} =: H_v^{\text{pre}}$, and $R_h = R_{h'} =: R_v$. The following backward construction computes the common payment responses and optimal policies:*

$$x_v^0(\eta) = [(r_v - a_v \eta)/q_v]_{[l_v, h_v]}, \quad (5)$$

$$d_v(\eta) = a_v x_v^0(\eta) + \beta \sum_w p_{vw} R_w(\eta), \quad (6)$$

$$R_v(\eta) = \min\{B_v, d_v(\eta)\}. \quad (7)$$

Write $d_v^- = a_v h_v + \beta \sum_w p_{vw} U_w$ for the finite constant on the left tail of d_v . If $B_v \geq d_v^-$, assign no barrier at v . Otherwise let α_v be the leftmost finite solution of $d_v(\alpha_v) = B_v$. The optimal local action and the incoming price sent to every successor are determined by

$$s_v = \max\{\eta, \alpha_v\}, \quad x_v = x_v^0(s_v), \quad (8)$$

where a missing barrier leaves $s_v = \eta$.

Every R_v is continuous, nonincreasing, piecewise affine with constant tails and range $[L_v, U_v]$. The global universe of distinct response knots has size at most $3N$. Complete response tables can be constructed in

$$A = O((N + M)N \log(N + 1)) \quad (9)$$

exact rational arithmetic operations and exact comparisons, using $O(N^2 + M)$ scalar storage. For a fixed root promise, choose any finite η_0 with $R_0(\eta_0) = b_0$. Execution uses at most $N + 1$ distinct incoming-price labels, drawn from η_0 and the finite cap barriers. The construction does not enumerate H occurrences.

Proof. The subtree below an occurrence is determined by its displayed vertex and subsequent public transitions. The map replacing the initial occurrence by another occurrence of the same vertex preserves all suffix paths, relative probabilities, discounts, local rewards, bounds, and caps. It is consequently a bijection of the normalized feasible problems at any common η . Strict concavity makes their maximizing action vectors agree. This proves the quotient independently of the algorithm.

Remove the current cap. Separability gives (5) and independent successor problems at the same incoming price, proving (6). Their unique maximizers have continuous payment responses. Backward induction starts with a clipped affine local payment and preserves continuity, monotonicity, and affine pieces under weighted summation. Its two tails are the extreme attainable payments before the current cap.

When the current cap is redundant, this pre-cap maximizer is feasible. Otherwise feasibility implies that B_v belongs to the range of d_v . The clipped finite response reaches its tails at finite prices, so a finite leftmost crossing exists. For $\eta < \alpha_v$, the pre-cap optimum at α_v has payment B_v . Let z^* be that optimum and z any policy satisfying the current cap. Writing G for conditional reward and Y for payment gives

$$G(z) - \eta Y(z) \leq G(z^*) - \alpha_v B_v + (\alpha_v - \eta) Y(z) \leq G(z^*) - \eta B_v.$$

For $\eta \geq \alpha_v$, the pre-cap optimizer already satisfies the cap. These two cases prove reflection, (7), and its implementing policy. If the payment is constant between two prices, the two optimality inequalities imply that either unique optimizer is also optimal at the other price. Thus the policy is unchanged on an equality plateau even when the supporting dual price is not unique. This also covers a cap at the minimum feasible payment and fixed local intervals.

Local clipping contributes at most two knots. Summing child responses introduces no knot outside their union. Capping can remove knots and introduces at most one crossing. Inducting over distinct *vertices*, rather than occurrences, gives a global union of at most $3N$. Each of the N responses can still have order N segments, so total response storage is quadratic, not linear. Directly merging weighted coefficient changes at sorted knots gives (9); the graph takes an additional $O(N + M)$ records.

Finally, each executed price is the maximum of the root price and the barriers along the realized path. There are at most $N + 1$ such numerical labels for this fixed root query, and at most $N(N + 1)$ reachable vertex–price pairs. The root maximizer at η_0 has precisely b_0 , so it also maximizes reward among policies satisfying the equality. $\square$

The unnormalized response at occurrence h in Proposition 2.1 is $w_h R_{v(h)}(\eta)$. This identity is the explicit link to an occurrence-indexed tree reduction. A normalized memoized implementation enjoys the same quotient recurrence; it is not an algorithmic competitor that our implementation is claimed to dominate.

3.2 Matching lower bounds for the response representation

The preceding upper bounds are not artifacts of a loose counting argument.

Theorem 3.2 (Worst-case tight response size). *For every integer $n \geq 1$, there is an n -vertex rational instance with $M = n - 1$ for which the global response-knot universe has exactly $2n$ distinct knots and the total number*

of affine response segments is exactly $n^2 + 2n$. Consequently the $O(N)$ global-knot bound and the $O(N^2 + M)$ complete-response storage bound in Theorem 3.1 are worst-case tight up to constant factors, even with no binding continuation caps.

Proof. Take the deterministic chain $1 \rightarrow 2 \rightarrow \dots \rightarrow n$, with $\beta = 1$, edge probability one, $a_i = q_i = 1$, $l_i = 0$, $h_i = 1$, and $r_i = 2i$. Set the continuation cap at vertex i to

$$B_i = n - i + 1.$$

This cap equals the maximum feasible remaining payment, so it is redundant and creates no barrier. The response at vertex i is therefore

$$R_i(\eta) = \sum_{j=i}^n [2j - \eta]_{[0,1]}. \quad (10)$$

For every j , the local summand has knots $2j - 1$ and $2j$. These $2n$ numbers are distinct. At each such number the slope of exactly one summand changes by one in magnitude, so no knot cancels in any ancestor response that contains that summand. Hence R_i has exactly $2(n - i + 1)$ knots and $2(n - i + 1) + 1$ affine segments. Summing over i gives

$$\sum_{i=1}^n [2(n - i + 1) + 1] = \sum_{k=1}^n (2k + 1) = n^2 + 2n.$$

All primitives are integers with $O(\log n)$ bits, so the family has polynomial encoding length. The claimed asymptotic lower bounds follow. $\square$

The theorem separates two notions that can otherwise be conflated. There are only linearly many distinct knot *locations*, but a downstream knot can appear in the response table of every ancestor, producing a quadratic number of stored affine pieces. The dense computational family in Section 10 is an empirical counterpart of this analytic construction; the lower bound itself does not depend on computation.

Corollary 3.3 (Several coordinates under one scalar commitment). *With d_v separable local coordinates, positive payment coefficients, positive quadratic coefficients, and a product interval, let $D = \sum_v d_v$. The knot universe has size at most $2D + N$, response storage is $O(N(2D + N) + M)$, and the incoming-price alphabet still has at most $N + 1$ elements for a fixed query.*

Proof. There are two clipping knots per coordinate, but only one scalar cap barrier per vertex. Clipping knots describe how a local action responds to a given price; they are not additional prices written into the running record. The common commitment price and cap reflection are unchanged. $\square$

Independent commitments can be separated only when rewards and feasible sets also separate. Coupled vector promises, nonseparable quadratic terms, and intertemporal switching do not satisfy the scalar reflection argument. Their retained primal promise-state and convex formulations remain applicable.

3.3 Rational bit complexity

An arithmetic count alone does not establish polynomial time on binary inputs. Let L be the total binary encoding length of all rational primitives, graph indices, and the rational root query, with rationals in lowest terms. The following statement is specifically for the coefficient-event construction used in Theorem 3.1. Stored rational coefficients, event locations, and crossings are reduced to lowest terms after each exact arithmetic update. The common denominators used in the proof below are bounding devices; they are not materialized as a giant implementation denominator.

Theorem 3.4 (Polynomial rational preprocessing for the coefficient-event compiler). *All stored response slopes, intercepts, finite knots, cap barriers, and a returned finite root price can be represented with*

$$S = O(NL + \log(N + M + 1)) \quad (11)$$

bits per reduced numerator and denominator. The event merges, sorting, exact comparisons, coefficient arithmetic, cap-crossing calculations, and one root inversion together admit the conservative bit-operation bound

$$O((N + M)N \log(N + 1) S^3) = O(AS^3). \quad (12)$$

The read-only response representation occupies $O((N^2 + M)S)$ bits. These are polynomial bounds for rational inputs to this explicit compiler, not a strongly polynomial claim and not a generic bit bound for every parametric optimization algorithm.

Proof. Fix an open interval between global knots. At a vertex whose current cap binds on that interval, the payment response is the constant B_v . Otherwise its coefficients are a local affine payment plus the discounted, probability-weighted coefficients at successors. The possible nonzero local sources are

$$B_v, \quad a_v l_v, \quad a_v h_v, \quad a_v r_v / q_v, \quad -a_v^2 / q_v. \quad (13)$$

Expanding either coefficient through the acyclic graph therefore gives a sum over directed paths of one source multiplied by edge probabilities and a power of β . Expansion stops at a capped vertex. No path repeats an edge, and its length is at most $N - 1$.

Let D_0 be the product of all positive denominators and the absolute values of all nonzero numerators appearing in the rational primitive input. Its logarithm is $O(L)$. Every path-term denominator divides D_0^{N+4} ; this exponent covers repeated discounting, each edge probability, and the products and divisions in (13). Thus a conceptual common denominator has $O(NL)$ bits even though the number of paths may be exponential. The sum of nonnegative discounted path weights to source occurrences is at most N , because it is bounded by the expected number of visited vertices. The absolute value of a source is at most $2^{O(L)}$. Multiplying by the conceptual denominator therefore bounds every reduced stored numerator and denominator by (11).

The actual event compiler does not build D_0^{N+4} . It adds and multiplies ordinary reduced rationals while merging sorted coefficient events. Intermediate stored partial sums have the same path-product denominator structure and differ by at most a polynomial factor in $N + M$ in magnitude. A new cap crossing has form

$(B_v - c)/m$ on a nonconstant line $m\eta + c$; local clipping knots and a root inverse are ratios of a constant number of bounded-height rationals. Their reduced heights also satisfy (11).

Sorting and merging are part of the operation count in (9). Every exact order test is a cross-product comparison of $O(S)$ -bit integers. Rational addition, multiplication, division, comparison, and gcd reduction can all be bounded conservatively by $O(S^3)$ bit operations. Multiplying that cost by the full arithmetic-and-comparison count proves (12). Counting stored coefficients, knot locations, and graph entries proves the storage claim. $\square$

The coefficient height above is not the bit length of a completely printed primal–dual certificate. Aggregate execution rewards and certificate totals can combine different realized denominators and have a larger, still polynomial, height. The Electronic Companion reports that separate aggregate bound and the implementation records coefficient precision separately from certificate precision. This distinction is part of the memory accounting below.

Corollary 3.5 (Amortized exact root inversion). *After the complete response family is compiled, a new feasible root promise can be inverted on the stored root response using $O(\log(N + 1))$ exact comparisons and a constant number of rational arithmetic operations. This is an inversion bound only: emitting and auditing the reached execution machine for the new query is a separate operation.*

Proof. The root response has at most $3N$ globally indexed knots. Binary search locates the constant or affine segment containing the requested payment; one affine solve, or selection of a finite point on a constant tail, returns a valid root price. $\square$

3.4 A compact exact certificate

For each reached (v, η) , propagate discounted occupation weight w , effective price s , action x , and multipliers $\chi = s - \eta$, $\ell, u \geq 0$. They satisfy

$$r_v - q_v x - a_v s + \ell - u = 0, \quad \ell(x - l_v) = u(h_v - x) = \chi(B_v - R_v(\eta)) = 0. \quad (14)$$

Weight $\beta w p_{vw}$ moves to (w, s) . A backward conditional-payment calculation verifies every cap; forward flow verifies occupation weights and the root promise. The resulting dual upper bound is

$$\eta_0 b_0 + \sum_{(v, \eta)} w \left[\chi B_v + u h_v - \ell l_v + \frac{(r_v - a_v s + \ell - u)^2}{2q_v} \right]. \quad (15)$$

Ancestor prices telescope under the flow equations, and (14) makes (15) equal the implemented reward. These rational checks certify a particular optimum; verifying entire response-curve identities is a separate implementation test. Neither kind of code execution replaces the mathematical proof above.

4 Piecewise-Quadratic Rewards and Shared Response Representations

The quadratic model identifies the basic mechanism, but a single curvature need not describe all service tiers. A change in staffing intensity can change marginal cost, and crossing a service threshold can produce a downward jump in marginal reward. Both features preserve exact closure. The relevant primitive is a finite, continuous, nonincreasing local price response, not a globally constant curvature.

4.1 Exact closure with changing curvature and marginal jumps

Replace the local reward in Section 2 by a continuous, strictly concave piecewise-quadratic function. More generally, vertex v may have d_v separable coordinates on a product interval, all contributing to the same scalar payment with coefficients $a_{vi} > 0$. For a nonfixed coordinate, let

$$l_{vi} = t_{vi,0} < t_{vi,1} < \dots < t_{vi,m_{vi}} = h_{vi}, \quad g_{vi}(x) = c_{vij} + r_{vij}x - \frac{1}{2}q_{vij}x^2 \quad (t_{vi,j-1} \leq x \leq t_{vi,j}),$$

where $q_{vij} > 0$. Adjacent formulas agree in value; the left derivative at a join is at least the right derivative. All breakpoints and coefficients are rational. A fixed coordinate contributes a constant and is recorded separately. Set $D = \sum_v d_v$, $J = \sum_{vi} m_{vi}$ over nonfixed coordinates, and $Q = 2J + N$. The transition, scalar commitment, and public cap assumptions remain those of Section 2.

Theorem 4.1 (Piecewise-quadratic quotient with a primitive event budget). *The local price response is*

$$\phi_{vi}(\eta) = \arg \max_{[l_{vi}, h_{vi}]} \{g_{vi}(x) - a_{vi}\eta x\}.$$

The occurrence quotient, payment recursion, and cap reflection remain exact, with

$$d_v(\eta) = \sum_i a_{vi}\phi_{vi}(\eta) + \beta \sum_w p_{vw}R_w(\eta), \quad R_v(\eta) = \min\{B_v, d_v(\eta)\}. \quad (16)$$

Every response is continuous and piecewise affine. At most Q globally distinct response knots occur. Explicit response tables use

$$O(NQ + J + D + M)$$

rational scalars and can be constructed using

$$A_{pq} = O((N + M)Q \log(Q + 1) + (J + D) \log(J + D + 2)) \quad (17)$$

exact arithmetic operations and comparisons. If all joins are differentiable, $2J$ in the knot budget can be replaced by $\sum_{vi} (m_{vi} + 1)$ over nonfixed coordinates.

For total binary input length L , a coefficient-event implementation with reduction after each rational update has response-coefficient and event height

$$S_{pq} = O((N + 1)L + \log(N + M + J + D + 2))$$

and bit-operation bound $O(A_{pq}S_{pq}^3)$. This bound includes sorting and comparison. For a fixed root promise, at most $N+1$ incoming-price labels are reached, independently of the number of local pieces. The behavioral minimization and price-order contiguity conclusions of Theorem 5.1 also remain valid.

Proof. On the interior of local piece j , the response is $(r_{vij} - a_{vi}\eta)/q_{vij}$. At a join t , the action equals t for every price between $g'_{vi,+}(t)/a_{vi}$ and $g'_{vi,-}(t)/a_{vi}$. These intervals have the correct orientation by concavity. The two domain endpoints give constant tails. Thus a coordinate with m pieces contributes at most two endpoint events and two events per internal join, or $2m$ in total. At a differentiable join the two events coincide. Strict concavity gives uniqueness, including at a nondifferentiable join; the local response is continuous and nonincreasing.

The suffix-path bijection in Theorem 3.1 does not depend on the reward being a single quadratic. Removing the current cap separates coordinates and successors, giving (16). The cap-reflection inequality in that theorem uses only optimality of the pre-cap continuation and linearity of its payment; it therefore applies without change. A weighted sum creates no knot outside the union of its inputs. A cap creates at most one new crossing. Induction over public vertices yields the global event budget. Merging weighted coefficient changes gives (17). Fixed coordinates contribute constants, not knots.

On an open interval between events, local payment sources now have the forms $a_{vi}t_{vi,j}$, $a_{vi}r_{vij}/q_{vij}$, and $-a_{vi}^2/q_{vij}$, together with cap constants. Expanding response coefficients along acyclic paths produces exactly the same path-product denominator structure as in Theorem 3.4. The product of primitive numerators and denominators has logarithm $O(L)$; raising it to $N + O(1)$ bounds the denominator of every path term. Summed absolute path weights times the local sources are bounded by a polynomial in $N + D$ times $2^{O(L)}$. Reduced partial sums therefore have height S_{pq} . Each crossing is a ratio of a constant number of bounded-height coefficients. Exact gcd reduction and cross-product comparison give the stated conservative Turing bound. These conceptual common denominators are not materialized by the implementation.

Only cap barriers change the carried price. Local marginal joins determine an action at that price but do not create a new writable price. Hence the running price is still the maximum of the root price and visited barriers. Along each suffix, every action coordinate is nonincreasing in the initial price; the endpoint-equality argument for contiguous behavioral classes remains applicable. Strict concavity preserves uniqueness and the exact-optimum randomized lower bound. $\square$

The extension includes genuine marginal discontinuities, not only a finer partition of an unchanged quadratic. It does not include affine pieces with zero curvature, coupled promises, or switching costs. Those changes can require different uniqueness or state arguments. The results above extend the exact theorem's domain without using those more general models to support an unproved closure claim.

4.2 A certificate that does not differentiate at a kink

Define the bounded local conjugate

$$k_v(s) = \sum_i \max_{x \in [l_{vi}, h_{vi}]} \{g_{vi}(x) - a_{vi}sx\}.$$

At each reached state (v, η) , let s be its reflected price, $\chi = s - \eta$, and $w_{v,\eta}$ its discounted occupation weight. Then

$$J(b_0) \leq \eta_0 b_0 + \sum_{(v,\eta)} w_{v,\eta} \{ \chi B_v + k_v(s) \}. \quad (18)$$

The compiled policy attains equality. Indeed, each local action attains its bounded conjugate, $\chi(B_v - Y_v) = 0$, and successor-price terms telescope under the occupation-flow equations. This is the same dual argument as Section 3.4, expressed without a derivative at a join or an artificial choice between adjacent quadratic formulas. All terms are rational at a rational compiled query. Aggregate certificate numerators may be larger than individual response coefficients; the coefficient-height bound is not an assertion that every printed certificate total has that same height.

4.3 Explicit tables versus a shared arithmetic circuit

The lower bound in Theorem 3.2 counts pieces in separately materialized vertex-response tables. It is not an information-theoretic lower bound for every exact representation of the family. That distinction leads to a complementary implementation result rather than a weaker size theorem.

Proposition 4.2 (Exact shared-circuit storage–query tradeoff). *After compilation, retain the local response curves, graph, cap barriers, and sorted global event set, but discard the per-vertex payment-response tables. The resulting shared circuit represents every R_v exactly using $O(J + D + N + M)$ rational scalars. For a common price, all vertex payments can be evaluated in*

$$T_{\text{eval}} = O \left(N + M + D + \sum_{vi} \log(m_{vi} + 1) \right)$$

arithmetic operations and comparisons, with the logarithmic sum over nonfixed coordinates. A new feasible root promise can be inverted using $O(T_{\text{eval}} \log(Q + 1))$ such operations. The result is a post-compilation storage bound; it does not assert that the preprocessing peak space is linear. A separately materialized class-transition table remains charged as in Theorem 5.1.

Proof. Store one addition-and-cap node per public vertex and one weighted link per edge. Local piecewise-affine evaluators have total size $O(J + D)$. At a supplied price, evaluate the circuit in reverse topological order using (16); shared successors are evaluated once. This gives the storage and evaluation counts. Binary search the globally sorted event locations using the circuit’s root payment. Between consecutive events the root response is affine, so evaluations at the two endpoints recover the exact inverse by linear interpolation in the payment–price plane. Constant intervals and tails return any finite supporting price, including a finite event endpoint. A constant root response needs no search. Cap barriers and the global event list have size $O(Q)$. $\square$

For the chain in Theorem 3.2, explicit tables have exactly $n^2 + 2n$ segments, whereas the shared circuit has n graph nodes and $3n$ local response segments. Both statements hold simultaneously. Materialization purchases faster repeated inversion; sharing saves read-only representation space. Neither representation

changes the extra writable alphabet of an already compiled exact policy. Thus read-only storage, combined execution states, and additional writable symbols are distinct design quantities even within the same problem instance.

4.4 An explicit bridge to parametric flow

The relation to parametric optimization can be stated constructively. In the occurrence tree of Proposition 2.1, introduce one vertex per occurrence and a sink. Send a feasible root supply $b \leq B_0$ through the tree. An arc from an occurrence's parent to that occurrence carries its total descendant allocation and has capacity $w_h B_{v(h)}$. An allocation arc from h to the sink carries z_h , has the transformed local bounds, and has cost

$$-w_h g_{v(h)}(z_h / (w_h a_{v(h)})).$$

Tree arcs have zero cost. Flow conservation makes their flows exactly the descendant sums. For multiple local coordinates, use parallel allocation arcs with their corresponding costs and coefficients. Hence this is a value-preserving convex-cost flow representation, including piecewise-quadratic rewards. Lower bounds can be shifted into node balances.

This reduction makes the comparison with Klimm and Warode (2021) precise: root supply is a primal demand parameter, whereas R_0 uses its dual price parameter; inversion connects the two. Applicability of a particular implementation must account for capacities, lower flows, and zero-cost structural arcs. The occurrence network can still be exponentially larger than the supplied public graph. The asserted contribution is therefore the primitive event budget and exact representation tradeoff on the quotient, not that piecewise-quadratic flow paths are new. The additional exact active-set reference described below computes complete parametric paths on small unfolded problems. It is not presented as an implementation or timing benchmark of the published parametric-flow algorithm.

5 Minimal Additional Writable Memory and Its Contractual Value

5.1 Observation architecture and the reached transducer

Fix the public graph, all primitives, and one root promise. At a decision, the decoder observes the current public vertex, including date, and a writable symbol. A read-only table may contain primitives, response coefficients, thresholds, the query's root price, and transition/output maps. Between decisions the update may depend on the old symbol and the realized public transition; the next action decoder has no separate access to the past transition. A retained promise, past tier, branch identity, or random seed must be charged as writable information if it carries history not contained in the current public vertex. A continuous remaining promise is therefore not an uncharged alternative register.

This architecture deliberately separates three resources. The *additional writable alphabet* is the changing record carried across recombination. The *combined reached behavior states* are the observable public vertex paired with a behavior class. The *read-only representation* contains the graph, compiled responses, coefficient

precision, and transition/output tables. None of these quantities is used as an unqualified notion of memory below. Changing the root promise requires a new inversion and can change the reached state set.

Let $\mathcal{P}_v$ be the incoming prices reachable at v under Theorem 3.1. For $\eta \in \mathcal{P}_v$, put $s_v(\eta) = \max\{\eta, \alpha_v\}$ with the missing-barrier convention. The reached system is a deterministic output transducer. Its state set is

$$Q = \{(v, \eta) : v \in \mathcal{V}, \eta \in \mathcal{P}_v\}.$$

At state (v, η) the output is $x_v^0(s_v(\eta))$. If the realized public transition is edge (v, w) , the next transducer state is $(w, s_v(\eta))$. The public edge is thus an exogenous input to the deterministic machine.

For two prices at the same public vertex, define behavioral equivalence backward:

$$\eta \sim_v \eta' \iff \begin{cases} x_v^0(s_v(\eta)) = x_v^0(s_v(\eta')), \\ s_v(\eta) \sim_w s_v(\eta') \text{ for every } (v, w) \in \mathcal{E}. \end{cases} \quad (19)$$

At a terminal vertex only the first condition is needed. This is the usual equality-of-output-and-successor-class recursion used in deterministic machine minimization. We use that classical principle explicitly (Bean et al., 1987; Givan et al., 2003; Peyrière, 2023); the new content lies in the contract-generated transducer and its ordered structure.

Theorem 5.1 (Minimal reached transducer and additional writable alphabet). *Let c_v be the number of classes of $\sim_v$ on $\mathcal{P}_v$. Under the preceding observation architecture:*

1. *the reached controller has*

$$C_* = \sum_{v \in \mathcal{V}} c_v \quad (20)$$

reachable public-state/class pairs;

2. *the minimum size of a common additional writable alphabet is*

$$K_* = \max_{v \in \mathcal{V}} c_v, \quad (21)$$

hence the changing record needs $\lceil \log_2 K_ \rceil$ bits;*

3. *every class is contiguous in the ordered list of incoming prices at its public vertex;*
4. *the backward signatures can be constructed on the reached vertex-price graph in $O((N+M)N \log(N+1))$ arithmetic comparisons and elementary index operations.*

The read-only response compiler remains charged separately: it stores $O(N^2 + M)$ rational scalars of coefficient height at most S from Theorem 3.4. A direct class-transition table has $\sum_v c_v \deg^+(v)$ entries. The exact-optimum lower bound for K_ also holds for randomized controllers whose retained private randomness is charged to the same writable state and whose continuation feasibility is imposed in conditional expectation at each public history.*

Proof. For a finite deterministic output transducer, two states can share one minimized state exactly when they emit the same current output and every common input leads to equivalent successors. Applied within a fixed observable public vertex, this is precisely (19). Backward partition refinement therefore yields the coarsest behavior-preserving partition of the reached price states. This is the classical minimization step.

Under the declared architecture the public vertex is observed without using the writable register. Thus each pair (v, class) is a distinct reached combined state, giving (20), while class indices may be reused across different public vertices. A common alphabet of size $\max_v c_v$ is sufficient: at vertex v , assign distinct labels only to its c_v classes, decode the common action, and update to the corresponding child label. Conversely, if two inequivalent prices at the same public vertex share one writable symbol, the deterministic decoder and update maps coincide from that point onward, so some positive-probability suffix receives a different action than the unique optimum. Hence at least c_v labels are necessary at each v , proving (21).

Along a fixed suffix path, the effective price is the maximum of the incoming price and a fixed finite set of barriers. Every local action is nonincreasing in that effective price. Thus every coordinate of the complete future plan is nonincreasing in the initial incoming price. If two ordered endpoint prices induce the same plan, every intervening price induces that same plan coordinate by coordinate. Hence each equivalence class is contiguous.

A signature consists of the local action and the tuple of successor-class indices. Processing signatures in reverse topological order over at most $N(N + 1)$ reached vertex-price pairs and at most $M(N + 1)$ pair transitions gives the stated conservative bound. The read-only counts are direct consequences of Theorems 3.1 and 3.4.

For a randomized feasible controller, take the mean action at each public history. Exogenous transitions and linear payments and caps make this mean vector feasible under conditional-expectation feasibility. Strict concavity gives

$$\sum_h w_h \mathbb{E}_{g_{v(h)}}(X_h) \leq \sum_h w_h g_{v(h)}(\mathbb{E}X_h) \leq J(b_0),$$

with equality only if each action is almost surely constant at the unique optimal value. Distinct required continuation plans must therefore have disjoint supports of retained symbols; a shared symbol with positive probability would impose the same future behavior on both histories. Additional retained randomness capable of distinguishing them is itself writable information under the stipulated accounting. At least c_v symbols are necessary. $\square$

The randomization convention has a concrete contractual interpretation. Conditional-expectation feasibility applies when participation is evaluated before an internal controller coin is realized and the private coin is not a contractually observable service state. If the institution instead requires every private randomization realization to satisfy every continuation cap pathwise, the admissible controller class is smaller. The deterministic optimum and the lower bound above remain valid; no smaller exact alphabet can be created by imposing the stronger feasibility requirement.

Corollary 5.2 (Linear writable-memory order is tight). *For every $k \geq 1$, the renewal family in Theorem 5.3 has $N = k + 2$ public vertices and satisfies $K_* = k = N - 2$. Hence the $O(N)$ upper bound on the additional writable alphabet is worst-case tight.*

Proof. The k intermediate branches recombine into one terminal public vertex and require k distinct optimal terminal actions. They are therefore k distinct behavioral classes at that terminal vertex. The root and terminal add two public vertices. $\square$

The theorem minimizes exact execution memory for this fixed optimum under a declared observation interface. It does not minimize the response compiler's storage or characterize the best approximate controller for every alphabet size. Those are separate design questions. The distinction is now explicit in K_* , C_* , and the read-only response size.

5.2 Ordered memory frontiers beyond equal branch probabilities

Consider three dates. The root tier is fixed at zero. At the intermediate date a branch j is observed with probability $\pi_j > 0$, where $\sum_j \pi_j = 1$. All branches then enter the same terminal public vertex. Payments and discounting equal one. Intermediate and terminal tiers are in $[0, 1]$. Branch j has cap $0 < b_1 < \dots < b_k < 1$, and the root promise is $\bar{b} = \sum_j \pi_j b_j$. At the intermediate date the reward is $-h_j(y)$, where h_j is convex, nondecreasing, and $h_j(0) = 0$. The common terminal reward $f(c)$ is continuous, concave, and strictly increasing on $[0, 1]$. The terminal cap is one. The terminal decoder observes only its public vertex and its memory symbol, not the inherited intermediate tier.

Theorem 5.3 (Heterogeneous renewal frontier). *The unique full-information policy is $y_j = 0$, $c_j = b_j$, with value $J = \sum_j \pi_j f(b_j)$. Exact implementation requires k symbols. With at most m deterministic terminal symbols, an optimal encoding has contiguous nonempty cells in cap order. For a cell $i, \dots, j$, its terminal tier is b_i and its loss is*

$$C(i, j) = \sum_{h=i}^j \pi_h [f(b_h) - f(b_i) + h_h(b_h - b_i)]. \quad (22)$$

The exact minimum loss is $D_m = D(\min(m, k), k)$, where

$$D(s, j) = \min_{s-1 \leq i < j} \{D(s-1, i) + C(i+1, j)\}, \quad D(0, 0) = 0, \quad (23)$$

and all other infeasible entries equal $+\infty$. With a precomputed cell-cost table this requires $O(mk^2)$ arithmetic operations. For quadratic f and quadratic h_j , prefix sums of the probability-weighted coefficients compute each cell cost in constant arithmetic time.

Proof. The root promise is the mean of the branch caps, so every branch cap must bind. Hence $y_j = b_j - c_j$ and $0 \leq c_j \leq b_j$. The branch reward $f(c) - h_j(b_j - c)$ is strictly increasing on that interval. Full information therefore uniquely selects $c_j = b_j$; different terminal actions require different symbols.

Within a memory cell all terminal actions coincide and cannot exceed its smallest cap. Monotonicity selects that cap and subtraction from the full-information reward gives (22). Fix any collection of cell minima. Every branch strictly prefers the largest eligible selected minimum, irrespective of its probability and intermediate cost. Reassigning branches in this way keeps each selected minimum in its own nonempty cell and produces contiguous cells without reducing value. Splitting at the final cell gives (23). Expanding the quadratic terms gives the stated prefix-sum implementation. $\square$

This strengthens and includes the quadratic equal-probability frontier. With $f(c) = 2c - c^2/2$, $h_j(y) = y^2/2$, and $\pi_j = 1/k$, (22) becomes

$$C(i, j) = \frac{2 - b_i}{k} \sum_{h=i}^j (b_h - b_i). \quad (24)$$

For $k = 8$ and $b_j = j/9$, one, two, and three symbols lose $119/162$, $5/18$, and $49/324$, respectively. Two writable bits permit four symbols, not three. The segmentation recurrence is standard; the contract-specific cell loss, ordered-cell sufficiency, and exact memory interpretation are the substantive statements.

The earlier nonsaturated two-branch example also remains useful. With branch caps $1/4$ and 1 , root promise $1/2$, and the same quadratic rewards, the full optimum uses terminal tiers $1/4$ and $3/4$ and has value $27/32$. An optimized public-only table uses common terminal tier $1/4$ and intermediate tiers $0, 1/2$, giving $13/32$. The adaptation gain is $7/16$ relative to the optimized restriction. At the full optimum both inherited intermediate tiers are zero, so making that tier observable does not evade this example's exact-memory lower bound. By contrast, the entire restricted-memory frontier above depends on not allowing a branch-specific intermediate tier to serve as an uncharged code.

5.3 Mean-preserving radial spreads of outside options

Fix positive probabilities, an interior mean $\bar{b}$, and strictly increasing numbers d_j with $\sum_j \pi_j d_j = 0$. Let

$$b_j(\varepsilon) = \bar{b} + \varepsilon d_j, \quad 0 < b_j(\varepsilon) < 1. \quad (25)$$

Keep the common service-benefit continuation fixed. The outside options are that continuation minus $b_j(\varepsilon)$. Increasing ε is therefore a mean-preserving radial spread of continuation caps and outside options along the fixed ordered direction d ; physical public transitions are unchanged.

Proposition 5.4 (Mean-preserving radial spreads increase the value of contractual memory). *For the quadratic renewal family in (24), with arbitrary positive branch probabilities and every strictly ordered zero-mean direction d , every insufficient deterministic alphabet $m < k$ has $D_m(\varepsilon) > 0$ and $D_m(\varepsilon)$ is strictly increasing in $\varepsilon > 0$ throughout the admissible range. Moreover $D_m(\varepsilon) = O(\varepsilon)$ as $\varepsilon \downarrow 0$. Exact execution needs one symbol at $\varepsilon = 0$ and k symbols at every $\varepsilon > 0$.*

Proof. For a fixed contiguous partition with cell minimum i , the loss on branch h in that cell is

$$\pi_h \varepsilon (d_h - d_i) (2 - \bar{b} - \varepsilon d_i).$$

For $h > i$, its derivative is

$$\pi_h (d_h - d_i) (2 - \bar{b} - 2\varepsilon d_i) > 0.$$

If $d_i \leq 0$ this is immediate. If $d_i > 0$, admissibility gives $\varepsilon d_i < 1 - \bar{b}$, so the last factor exceeds $\bar{b} > 0$. A partition with fewer than k cells has at least one positive term. Every candidate partition cost is therefore strictly increasing and positive. The minimum of finitely many strictly increasing candidate costs is strictly

increasing: evaluate at the minimizing partition for the larger parameter. Each cost is $O(\varepsilon)$ near zero. Distinct optimal terminal tiers give the alphabet statement. $\square$

The proposition is intentionally a theorem about every ordered radial spread direction, not a claim about arbitrary convex-order, majorization, or variance comparisons between unrelated cap vectors. It is nevertheless stronger than an equal-weight or one-calibration statement: branch probabilities are arbitrary positive numbers and the direction d is arbitrary subject to strict order and zero weighted mean. The economic distinction remains sharp: the exact alphabet can jump discontinuously at zero spread while the value loss from insufficient memory converges continuously to zero.

6 When Randomization Improves a Restricted Memory Budget

The exact-optimum lower bound in Theorem 5.1 does not imply that randomization is useless below that alphabet size. The renewal family permits an explicit answer under the declared conditional-expectation participation institution. At an intermediate public branch, the controller may randomize its terminal symbol; it carries no uncharged seed or branch record. Participation is evaluated before that private draw. Terminal actions may depend only on the terminal public vertex and the retained symbol.

For terminal levels $c_1 < \dots < c_s$, let F_c be the linear interpolant through $(c_i, f(c_i))$. Since f is concave and strictly increasing, this interpolant is concave and strictly increasing on $[c_1, c_s]$; for $s = 1$ it is defined at its sole point. Retain the caps, probabilities, and convex intermediate costs of Theorem 5.3.

Theorem 6.1 (Optimal lotteries for a fixed terminal alphabet). *For a fixed set of terminal levels, a feasible randomized controller exists if and only if $c_1 \leq b_1$. Its optimum at branch j has deterministic intermediate tier and terminal mean*

$$\mu_j = \min\{b_j, c_s\}, \quad y_j = b_j - \mu_j.$$

It randomizes over at most the two adjacent terminal levels bracketing μ_j . The optimal loss with at most m writable terminal symbols is consequently

$$D_m^{\text{ran}} = \min_{1 \leq s \leq m, \substack{0 \leq c_1 < \dots < c_s \leq 1 \\ c_1 \leq b_1}} \sum_j \pi_j \{f(b_j) - F_c(\min\{b_j, c_s\}) + h_j((b_j - c_s)_+)\}. \quad (26)$$

The minimum allows coincident levels by deleting unused duplicates. It satisfies $D_m^{\text{ran}} \leq D_m$. Equality with the full-information optimum still requires k symbols when the full-information actions are distinct and rewards are strictly concave. Formula (26) optimizes the encoding for each continuous codebook; it is not a claim that the joint codebook problem is solved by the deterministic segmentation recurrence.

Proof. The root promise equals the weighted mean of the branch caps. Every branch cap must therefore bind in conditional expectation, so $\mathbb{E}(Y_j + C_j) = b_j$. Replace a randomized terminal decoder, conditional on a symbol, by its mean. It uses the same number of symbols, preserves payments, and weakly increases reward by concavity. Replace a randomized intermediate tier by its branch mean; convexity of h_j weakly lowers its cost without altering the encoder's terminal-symbol probabilities. These replacements use exogenous transitions and the absence of switching costs.

For a terminal mean μ in the convex hull of the levels, the largest expected terminal reward is $F_c(\mu)$. The upper hull of the concave points $(c_i, f(c_i))$ consists of adjacent chords; a lottery over their endpoints attains the hull. The branch objective is $F_c(\mu) - h_j(b_j - \mu)$ on $[c_1, \min(b_j, c_s)]$. It is strictly increasing, so its maximum occurs at μ_j . Feasibility requires $c_1 \leq b_j$ for every branch, equivalently $c_1 \leq b_1$. Subtraction from full information gives the formula. Compactness of symbol probabilities and levels gives attainment, with redundant symbols deleted. Deterministic encodings are admissible randomized encodings, yielding the inequality. The exact-optimum statement follows from the strict-concavity argument in Theorem 5.1, not from the restricted-alphabet formula alone. $\square$

Proposition 6.2 (A strict, exactly solved two-symbol separation). *In the quadratic renewal model, take three equiprobable branches with caps $1/4, 1/2, 3/4$, $f(c) = 2c - c^2/2$, and $h_j(y) = y^2/2$. Then*

$$D_2 = \frac{1}{8}, \quad D_2^{\text{ran}} = \frac{1}{96}, \quad D_3 = D_3^{\text{ran}} = 0. \quad (27)$$

Two symbols thus reduce the randomized loss to one twelfth of the best deterministic loss, without reducing the three-symbol requirement for exact implementation.

Proof. The two deterministic contiguous partitions have losses $7/48$ and $1/8$. For the randomized problem, let the two levels be $u \leq v$. Any choice with $v < 1/4$ can be improved by raising its levels. For $v \geq 1/4$, feasibility requires $u \leq 1/4$; raising u to $1/4$ weakly increases every relevant concave chord value. Thus take $u = 1/4$.

For v in $[1/4, 1/2]$, $[1/2, 3/4]$, and $[3/4, 1]$, respectively, Theorem 6.1 gives loss

$$\begin{cases} [(1/2 - v)(2 - v) + (3/4 - v)(2 - v)]/3, \\ [(v - 1/2)/8 + (3/4 - v)(2 - v)]/3, \\ [(v - 1/2)/4 + (v - 3/4)/2]/6. \end{cases}$$

The first two expressions strictly decrease on their domains and the third strictly increases. Hence the global minimum is at $v = 3/4$ and equals $1/96$. The first and third branches use their respective endpoint symbols; the middle branch uses each with probability one half. All intermediate tiers are zero. The middle branch's loss is the strict-concavity loss $1/32$, weighted by its probability $1/3$. Three distinct terminal symbols attain full information. $\square$

The institutional qualification is substantive. On the middle branch, the high private draw delivers terminal payment $3/4$ against an expected-continuation cap $1/2$. The lottery is feasible when participation is checked before the private draw, as stipulated here, but not when that branch cap must hold for every private draw. We do not transfer the randomized frontier to the pathwise institution. The deterministic frontier and the exact-memory lower bound retain their original domains. This separation explains why a statement about the minimum alphabet at the exact optimum cannot settle the economic value of randomization under a smaller alphabet.

7 An Exact Global Randomized Memory Frontier

The fixed-codebook formula in Theorem 6.1 leaves a continuous design problem: both the terminal service levels and their assignment probabilities must be chosen. We now solve that problem globally for quadratic terminal rewards and heterogeneous quadratic intermediate costs. The solution also determines when imposing participation after, rather than before, the private draw eliminates the value of randomization. These are restricted-alphabet design results, distinct from the exact-implementation alphabet in Theorem 5.1.

Retain the three-date renewal architecture of Theorem 5.3, including the saturated root promise $\bar{b} = \sum_{j=1}^k \pi_j b_j$, strictly ordered caps $0 < b_1 < \dots < b_k < 1$, and positive probabilities summing to one. Throughout this section,

$$f(c) = rc - \frac{q}{2}c^2, \quad h_j(y) = \frac{\gamma_j}{2}y^2, \quad r \geq q > 0, \quad \gamma_j \geq 0. \quad (28)$$

The parameters and probabilities can differ across branches as indicated. Terminal reward is strictly increasing on the feasible cap range, including when $r = q$. All private information retained until the common terminal vertex must be encoded in the charged symbol. The decoder and public observation convention are unchanged.

7.1 Cap anchors and one continuous terminal level

For $u \leq b \leq v$, quadratic concavity gives the interpolation loss

$$f(b) - \frac{v-b}{v-u}f(u) - \frac{b-u}{v-u}f(v) = \frac{q}{2}(b-u)(v-b), \quad (29)$$

with the endpoint value understood when levels coincide. Thus an interior codeword enters the loss affinely until it crosses a branch cap. The highest codeword is different: reducing it below a branch cap creates an intermediate payment and its associated cost.

Lemma 7.1 (Cap-anchored representation). *For every alphabet budget $m \geq 1$, an optimum of (26) exists with the following representation. Its lowest codeword is b_1 ; every codeword except the highest belongs to $\{b_1, \dots, b_k\}$; and the highest belongs to $[b_1, b_k]$. A one-symbol optimum is $\{b_1\}$. The highest codeword need not be a branch cap.*

Proof. Attainment follows from the compact level-and-probability formulation in Theorem 6.1. Delete unused levels and coincident duplicates. If all levels lie below b_1 , raising the largest to b_1 improves every branch and leaves only that level relevant. Otherwise delete levels below b_1 except the largest such level, if any, and raise it to b_1 . Every affected cap is bracketed by this level and the next one. Equation (29) shows that raising the lower endpoint weakly lowers loss. This also covers a one-level codebook. The same argument at the other extreme deletes superfluous levels above b_k and lowers the remaining highest endpoint to b_k without increasing loss.

Fix the highest level. Take a nonhighest codeword not equal to a cap. Between the nearest caps and its neighboring codewords, the set of branch caps in each adjacent interpolation interval is fixed. Equation (29) makes the sum of their losses affine in the moving codeword; all other losses are unchanged. Move it to an

endpoint that does not increase this affine function. At a cap it becomes anchored. At a collision it can be deleted: no strictly intervening branch cap has been crossed, and the limiting interpolation losses agree. Each move reduces the number of unanchored nonhighest levels or the number of levels. Finitely many moves produce the stated representation. Branches above the fixed highest level are unaffected throughout. $\square$

The lemma does not discretize the highest level. A cap-only restriction can be strictly suboptimal, as Proposition 7.3 shows. It also does not assert that every optimal codebook is anchored; it supplies an optimum with this useful form.

7.2 A finite exact algorithm

For adjacent anchored levels $b_i < b_j$, define

$$E(i, j) = \frac{q}{2} \sum_{h=i}^j \pi_h (b_h - b_i)(b_j - b_h). \quad (30)$$

Let $P(s, j)$ be the minimum interpolation loss on caps through b_j using exactly s anchored levels, starting at b_1 and ending at b_j . Set $P(1, 1) = 0$, $P(1, j) = +\infty$ for $j > 1$, and use the ordered recurrence

$$P(s, j) = \min_{i < j} \{P(s-1, i) + E(i, j)\}, \quad 2 \leq s \leq j. \quad (31)$$

This is a standard shortest-path or segmentation recurrence. The substantive reduction is Lemma 7.1, which makes it applicable to a jointly optimized continuous lottery codebook.

Suppose that b_i is the last anchored level and z is the final level. For $\ell \geq i$ and $b_\ell \leq z \leq b_{\ell+1}$, put $b_{k+1} = b_k$ and define

$$\begin{aligned} \Psi_{i\ell}(z) &= \frac{q}{2} \sum_{h=i+1}^{\ell} \pi_h (b_h - b_i)(z - b_h) \\ &\quad + \sum_{h=\ell+1}^k \pi_h \left\{ f(b_h) - f(z) + \frac{\gamma_h}{2} (b_h - z)^2 \right\}. \end{aligned} \quad (32)$$

The first sum is the lottery loss below z ; the second is the terminal-reward and intermediate-payment loss above it. Empty sums are zero. For $\ell < k$ the function is strictly convex in z and has the constrained minimizer

$$z_{i\ell} = \left[\frac{r \sum_{h>\ell} \pi_h + \sum_{h>\ell} \pi_h \gamma_h b_h - \frac{q}{2} \sum_{h=i+1}^{\ell} \pi_h (b_h - b_i)}{q \sum_{h>\ell} \pi_h + \sum_{h>\ell} \pi_h \gamma_h} \right]_{[b_\ell, b_{\ell+1}]}. \quad (33)$$

For $\ell = k$, take $z_{ik} = b_k$. Define the terminal cost

$$T_i = \min_{i \leq \ell \leq k} \Psi_{i\ell}(z_{i\ell}). \quad (34)$$

Theorem 7.2 (Global frontier and exact complexity). *Under (28), the optimum with at most m symbols is*

$$D_m^{\text{ran}} = \min \left\{ D_1, \min_{\substack{1 \leq s \leq \min(m-1, k) \\ 1 \leq j \leq k}} \{P(s, j) + T_j\} \right\}, \quad (35)$$

where the inner minimum is empty for $m = 1$. Backtracking produces a globally optimal codebook and the adjacent lotteries implementing it. For ordered inputs and $1 \leq m \leq k$, all frontiers through budget m can be computed in $O(mk^2)$ exact rational operations and $O(mk + k)$ stored rational numbers and indices, including backpointers. No codeword grid is required.

For rational input of total binary length L , a reduced-rational implementation has polynomial Turing complexity. A conservative bound is $O(mk^2 S^3)$ bit operations with $S = O(k[L + \log(k + 2)])$. This includes exact cost comparisons, clamping, and fraction reduction. For $m \geq k$ the optimal loss is zero; listing additional budget entries requires only their output cost. These guarantees concern the stated algorithm, not an arbitrary nonlinear optimizer for (26).

Proof. By Lemma 7.1, a nontrivial optimal codebook consists of an anchored prefix and one final level. The prefix contributes exactly the sum of its edge costs. Splitting at the preceding anchor proves (31) by induction on the number of anchors. If two successive edges share an endpoint cap, its loss is zero, so no term is double counted. Conditional on the final anchor b_j , every possible highest level belongs to one of the intervals in (32). Differentiating that quadratic gives (33); its quadratic coefficient is strictly positive whenever $\ell < k$ because the remaining probabilities and q are positive. Formula (34) therefore optimizes over the entire remaining continuous domain.

Every term in (35) reconstructs a feasible controller with at most m codewords. A coincident final level is deleted. Conversely, the representation lemma maps an optimal controller into one of those terms, including the one-symbol alternative. The two inequalities prove equality, not merely an upper bound or a stationary-point condition. Recover the final level attaining T_j , backtrack the minimizing indices in P , and apply Theorem 6.1 to obtain the encoder probabilities and intermediate tiers.

Prefix sums of π_h , $\pi_h b_h$, $\pi_h b_h^2$, $\pi_h \gamma_h$, $\pi_h \gamma_h b_h$, and $\pi_h \gamma_h b_h^2$ evaluate each edge and each quadratic tail in constant arithmetic work. All T_i require $O(k^2)$ work. The prefix dynamic program requires $O(mk^2)$ work and $O(mk)$ backpointers; edge costs are evaluated on demand rather than stored in a quadratic array. The six moment arrays and all terminal optima require $O(k)$ storage.

For completeness, each moment, edge coefficient, and tail coefficient is a sum of bounded-degree monomials in rational inputs. A conceptual common denominator is a fixed power of the product of input denominators; it has polynomial bit length and is not constructed by the implementation. Each terminal minimizer adds one division of such coefficients. Each prefix cost adds at most k edges; each final candidate adds one tail optimum. Reducing after every operation gives the conservative height bound S . Exact comparisons use cross multiplication on integers of $O(S)$ bits, and ordinary integer arithmetic with greatest-common-divisor reduction costs at most $O(S^3)$ per rational operation. This establishes the stated bound, including comparisons rather than only arithmetic counts. The electronic companion gives an implementation-level concordance. $\square$

7.3 A genuinely continuous optimal codebook

Proposition 7.3 (An off-cap terminal level is optimal). *Let $(b_1, b_2, b_3) = (1/4, 1/2, 3/4)$, $(\pi_1, \pi_2, \pi_3) = (7/20, 3/5, 1/20)$, $r = 2$, $q = 1$, and $\gamma_j = 1$. The globally optimal two-symbol expected-participation codebook is $(1/4, 5/8)$, and*

$$D_2^{\text{ran}} = \frac{23}{1280} < \frac{3}{160} = \min_{\substack{|c| \leq 2 \\ c \subseteq \{b_1, b_2, b_3\}}} D(c) = D_2, \quad (36)$$

where the cap-only minimum is over feasible codebooks and $D(c)$ denotes their expected-participation loss. Thus even optimizing all cap-only lotteries misses the unrestricted optimum.

Proof. The representation lemma fixes the lower level at $1/4$. For an upper level $z \in [1/2, 3/4]$, the loss is

$$\frac{3}{40}(z - 1/2) + \frac{1}{20}(3/4 - z)(2 - z) = \frac{z^2}{20} - \frac{z}{16} + \frac{3}{80}.$$

Its unique minimum is at $z = 5/8$ and equals $23/1280$. On $[1/4, 1/2]$, both higher-cap branches lie above z and the loss strictly decreases up to $1/2$. Levels above $3/4$ can be lowered, as in Lemma 7.1. These observations also exclude the one-level boundary. The cap-only choices with two feasible levels have losses $3/160$ and $3/160$; one level has loss $49/160$. The deterministic partition using levels $1/4, 1/2$ has loss $3/160$, while the other contiguous two-cell partition has loss $21/80$. Theorem 5.3 yields the deterministic optimum.

The middle branch draws terminal level $5/8$ with probability $2/3$ and level $1/4$ with probability $1/3$, with zero intermediate tier. The highest-cap branch uses intermediate tier $1/8$ and terminal tier $5/8$. All branch payments equal their caps in expectation, and the root payment is $17/40$. The resulting weighted losses are $3/320$ and $11/1280$, respectively, summing to $23/1280$. $\square$

7.4 Participation after the private draw

The preceding frontier is appropriate when a renewal review approves an expected continuation before the protocol draws its private symbol. Consider instead an agreement that requires the same cap to hold for every private realization after that review. A draw-specific intermediate tier is allowed; only the terminal public state and the charged symbol reach the terminal decoder. Write $D_m^{\text{pw,ran}}$ for the best randomized loss under these pathwise constraints and the same saturated root promise.

Theorem 7.4 (Pathwise randomization has no frontier advantage). *For the general renewal family of Theorem 5.3, with strictly increasing concave f and convex nondecreasing h_j , and the observation architecture just specified,*

$$D_m^{\text{pw,ran}} = D_m \quad \text{for every } m \geq 1. \quad (37)$$

Consequently (35) and the deterministic frontier give an exact comparison of the two participation institutions under (28). Their exact-optimum alphabet thresholds are both k when strict concavity makes the k distinct full-information actions unique.

Proof. Pathwise feasibility gives $Y_j + C_j \leq b_j$ almost surely at each branch. The root promise is $\sum_j \pi_j b_j$ and every branch has positive probability. Hence $Y_j + C_j = b_j$ almost surely on every branch: any positive expected slack would contradict root saturation.

A retained symbol determines a terminal level, after replacing an independent randomized terminal decoder by its mean. This replacement preserves feasibility and weakly improves reward. To see this under pathwise saturation, conditional on a branch and symbol the only permissible additional terminal randomization is fresh and independent of the earlier tier; otherwise a retained shared seed would be an uncharged state. Independence together with $Y_j + C_j = b_j$ almost surely forces that decoder to be degenerate. This also shows why the memory architecture matters.

For a fixed codebook, a symbol with terminal level c can be used on branch j only if $c \leq b_j$; saturation fixes its intermediate tier at $b_j - c$. Its payoff is $f(c) - h_j(b_j - c)$, which is strictly increasing in c . Thus the highest eligible codeword weakly dominates every lottery over eligible symbols, including lotteries with draw-specific intermediate tiers. Selecting it deterministically preserves each branch's payment and cannot reduce payoff. Optimizing codebooks yields the deterministic ordered frontier of Theorem 5.3. The reverse inequality follows because every deterministic feasible controller is an admissible pathwise-randomized controller. Strict-concavity uniqueness gives the final threshold assertion. $\square$

This institutional result is not an assertion that randomization is generally ineffective in dynamic contracts. It uses the saturated root promise, an exogenous recombination, and the charged-information interface. Under expected participation, Theorem 7.2 chooses a less expensive lottery protocol; under pathwise participation, the same service model calls for the deterministic frontier. In either institution an operator can price a symbol budget, add its implementation charge to the computed loss, and optimize the budget without confusing read-only program size with retained private information.

8 Monge Structure and Exact Frontier Acceleration

The global construction in Section 7 searches every terminal interval for every last anchor and every preceding anchor for every prefix state. Neither quadratic scan is necessary. We show that both search families have ordered minimizers, including after the continuous highest level has been optimized. The deterministic frontier has the same property. This provides a same-input acceleration of the complete institutional comparison, rather than a speed ratio against an unfolded problem.

The matrix-search principle is classical. Quadrangle inequalities and monotone search have long been used to accelerate dynamic programs (Yao, 1980; Aggarwal et al., 1987). Our contribution here is to establish that structure for the contractual interpolation and *continuously minimized terminal* costs. The implementation below uses elementary divide-and-conquer search; it does not claim to implement the stronger linear-search bounds available for suitable totally monotone matrix interfaces.

Let $I_\ell = [b_\ell, b_{\ell+1}]$ for $\ell < k$, and $I_k = \{b_k\}$. Define

$$Q(i, \ell) = \min_{z \in I_\ell} \Psi_{i\ell}(z), \quad i \leq \ell \leq k. \quad (38)$$

Thus $T_i = \min_{\ell \geq i} Q(i, \ell)$. A cost array is Monge on its feasible domain if every feasible ordered two-by-two subarray satisfies the uncrossing inequality: its northwest plus southeast entries do not exceed its northeast plus southwest entries. We always select the leftmost minimizer.

Theorem 8.1 (Monge frontiers). *Under (28), the following statements hold.*

1. *The anchored-edge costs E and deterministic cell costs C satisfy*

$$A(a, c) + A(b, d) \leq A(a, d) + A(b, c), \quad a \leq b \leq c \leq d, \quad (39)$$

for $A = E, C$. The cell-cost statement also holds under the general assumptions of Theorem 5.3.

2. *The interval-minimized terminal array Q is Monge on $i \leq \ell$. Its leftmost minimizing interval is nondecreasing in the final-anchor index. The leftmost preceding-anchor indices in every feasible layer of (31) and (23) are nondecreasing in the endpoint index.*
3. *For ordered rational inputs and $1 \leq m \leq k$, both complete frontiers through budget m , including optimal codebooks, can be computed in*

$$O(mk \log(k+1)) \quad (40)$$

exact rational operations and comparisons, using $O(mk + k)$ stored scalars and indices. With the coefficient-height bound S in Theorem 7.2, a conservative Turing bound is $O(mk \log(k+1)S^3)$. This changes computation, not the feasible institution or the optimal frontier.

Proof. Expanding the quadratic interpolation costs gives (39) for E : the difference between its right and left sides is a sum of nonnegative cap gaps times positive probabilities. For C , that difference is the sum over $h = c + 1, \dots, d$ of

$$\pi_h \{f(b_b) - f(b_a) + h_h(b_h - b_a) - h_h(b_h - b_b)\},$$

which is nonnegative. The companion supplies the explicit edge factorization, including coincident-index boundaries.

The terminal minimization requires an additional argument. Write $\Psi_i(z)$ for the piecewise function obtained by joining (32). For $i < j$ and $z \geq b_j$, cancellation of all above- z costs gives

$$\begin{aligned} \Psi_i(z) - \Psi_j(z) = \frac{q}{2} \left[\sum_{h=i+1}^j \pi_h(b_h - b_i)(z - b_h) \right. \\ \left. + (b_j - b_i) \sum_{b_j < b_h \leq z} \pi_h(z - b_h) \right]. \end{aligned} \quad (41)$$

This continuous function is nondecreasing in z . Consequently $\Psi_i(x) + \Psi_j(y) \leq \Psi_i(y) + \Psi_j(x)$ whenever $b_j \leq x \leq y$. For two ordered terminal intervals, choose the minimizers of the two crossed entries of Q . They are ordered because the intervals are ordered. Applying the preceding inequality and then minimizing

the uncrossed entries proves the Monge inequality for Q . Merely knowing that each terminal cell is convex would not establish this conclusion.

Adding a preceding-layer cost to each column preserves the Monge inequality. Its standard exchange argument makes leftmost row minimizers nondecreasing. The triangular feasible domains cause no exception: in a hypothetical crossing, the two crossed indices are feasible for both rows. A midpoint-row search may therefore restrict its left descendants to columns no larger than its minimizer, and its right descendants to columns no smaller. At each recursion depth the scanned intervals overlap only at their boundaries. This gives $O(k \log(k + 1))$ oracle calls per layer and for all terminal minima. Each oracle uses the six moment arrays from Theorem 7.2, including exact terminal clamping. The companion gives the search invariant and output accounting.

The frontier combination needs $O(mk)$ further comparisons. Reconstructing only the winning codebook once per budget takes $O(m^2)$ work and storage, already within $O(mk)$ for $m \leq k$. Two cost rows, terminal optima, and retained backpointers give the stated storage. Search changes which entries are visited but not their algebraic construction; therefore the same reduced-rational height bound applies. Exact comparison and fraction reduction give the stated Turing bound. $\square$

The economic comparison is unchanged: conditional-expectation participation uses the randomized frontier, while private-draw pathwise participation uses the deterministic frontier by Theorem 7.4. Heterogeneous intermediate curvatures cancel in (41), so they do not destroy terminal monotonicity. Convex nondecreasing nonquadratic intermediate costs retain that monotonicity with a one-dimensional minimization oracle; the exact rational work bound here remains specific to the quadratic input interface.

8.1 A linear-work implementation of both frontiers

Monotone row minima alone do not justify linear matrix search: the property must hold in every submatrix, including those with no feasible entry in a row. We now supply that interface for the contractual costs. The search algorithm is the classical SMAWK method of Aggarwal et al. (1987); the construction below establishes its applicability, not a new general matrix-search principle.

Theorem 8.2 (Linear-work contractual frontiers). *Under (28), with ordered rational inputs and $1 \leq m \leq k$, the complete expected-participation and deterministic/pathwise frontiers through budget m , together with one optimal codebook per budget, can each be computed in $O(mk)$ exact rational operations and comparisons and $O(mk + k)$ stored scalars and indices. With the coefficient height S of Theorem 7.2, a conservative bit bound is $O(mkS^3)$. The highest randomized level is optimized over its continuous interval, not restricted to a grid or to the caps.*

Proof. Give a feasible entry in column c the lexicographic key $(0, A_{rc}, c)$, so finite ties go left. For the terminal array, give an infeasible prefix entry $c < r$ the key $(1, 0, -c)$; for a prefix-program row, give an infeasible suffix entry the key $(1, 0, c)$. All feasible keys precede every infeasible key. Discard globally unreachable predecessor columns before completing a program layer.

These completed matrices are totally monotone. Indeed, for $r < s$ and $c < d$, strict preference for d over c in row r persists in row s . When all four entries are feasible, Theorem 8.1 implies this assertion; adding the

preceding value function to each column preserves the inequality. In the terminal staircase, later rows can lose feasible columns only from the left. If both compared entries become infeasible, the negative column index continues to prefer d . In a program layer, feasible prefixes expand with the row index: an earlier preference for d requires both compared columns to be feasible, so both remain feasible. The argument applies to arbitrary submatrices, not only consecutive rows or columns.

SMAWK therefore finds all terminal minima and each program layer in $O(k)$ key comparisons. Each finite key uses the constant-work exact moment oracle; padded keys never evaluate an economic cost. Scanning final-anchor candidates adds $O(mk)$ work. Reconstructing one winning codebook per budget adds $O(m^2) \leq O(mk)$. The search stores linear index lists, not a quadratic matrix; two value rows, terminal optima, and backpointers give the stated space. The arithmetic expressions are unchanged, so exact fraction reduction and comparison retain the stated coefficient-height and bit bounds. The companion records the reduction and interpolation invariants. $\square$

A uniform numerical infinity with ordinary leftmost ties is not a substitute for this completion. On the upper staircase, columns 1 and 2 can prefer 2 at row 2 but prefer 1 when both become infeasible at row 3. The ordinal completion resolves this submatrix obstruction without altering any feasible optimum. It improves the proved work bound, not the institution, the memory accounting, or the admissible cost class.

9 The Optimized Shared-Table Comparator

A prescribed observation restriction is optimized before its adaptation value is assessed. Let $c(v) \in \{1, \dots, C_{\text{info}}\}$ be its information cell and let u_c be a single table chosen before uncertainty. The restricted value is denoted K^{tab} , keeping the number of cells distinct from the value.

Proposition 9.1 (Zero-release graph formulation). *The optimized zero-release comparator is the convex quadratic program*

$$\begin{aligned} K^{\text{tab}} &= \max_{u, y} \sum_v \bar{w}_v g_v(u_{c(v)}) \\ y_v &= a_v u_{c(v)} + \beta \sum_w p_{vw} y_w, \quad y_v \leq B_v, \quad y_0 = b_0, \\ l_v &\leq u_{c(v)} \leq h_v. \end{aligned} \tag{42}$$

Here $\bar{w}_v = \beta^{t(v)} \Pr(v)$ is the graph's forward reach weight. There are $C_{\text{info}} + N$ variables and $O(N + M + C_{\text{info}})$ coefficient occurrences. For a fixed shared table, an incoming absolute switching cost can also be represented exactly by its root term and one reach-weighted epigraph term per public edge.

Proof. A common table fixes every local action by the public vertex. Backward conditional expectation gives the displayed y_v at every occurrence of v . Thus all history caps are equivalent to the N displayed caps. Conversely, every feasible table and payment vector implements accepted actions at all histories. Summing rewards by public reach weights proves the objective identity. An edge uses two fixed table entries, so its switching cost is also determined without additional history. $\square$

This supporting formulation uses standard convex optimization; it is not the central algorithmic novelty. In particular, a public table is not the full-information policy class. The difference $J(b_0) - K^{\text{tab}}$ compares two optimized protocols on identical primitives.

At a fixed table and release $\delta \geq 0$, permit $|x_h - u_{c(v(h))}| \leq \delta \omega_{c(v(h))}$, where $\omega_c \geq 0$. Set

$$l_v^{\text{eff}} = \max(l_v, u_{c(v)} - \delta \omega_{c(v)}), \quad h_v^{\text{eff}} = \min(h_v, u_{c(v)} + \delta \omega_{c(v)}).$$

Empty intervals or an infeasible root promise are rejected. Otherwise the finite-price compiler solves this fixed query exactly.

Proposition 9.2 (Supporting cuts including coincident bounds). *At every feasible fixed query (u, δ) , physical and corridor multipliers can be split so that nonnegative corridor multipliers $\rho_{v,\eta}^-, \rho_{v,\eta}^+$ give*

$$G_c = \sum_{(v,\eta):c(v)=c} w_{v,\eta}(\rho_{v,\eta}^+ - \rho_{v,\eta}^-), \quad H_\delta = \sum_{(v,\eta)} w_{v,\eta} \omega_{c(v)}(\rho_{v,\eta}^+ + \rho_{v,\eta}^-). \quad (43)$$

For any other feasible design $(\tilde{u}, \tilde{\delta})$,

$$V(\tilde{u}, \tilde{\delta}) \leq V(u, \delta) + G^\top (\tilde{u} - u) + H_\delta (\tilde{\delta} - \delta). \quad (44)$$

This statement includes physical/corridor ties and zero-width intersections.

Proof. At a reached pair let $d = r_v - q_v x - a_v s$. Choose effective lower and upper multipliers $\ell = \max(-d, 0)$, $u^+ = \max(d, 0)$. A positive lower multiplier occurs only when $x = l_v^{\text{eff}}$; a positive upper multiplier only when $x = h_v^{\text{eff}}$. Among the original lower inequalities, at least one attains l_v^{eff} . Assign nonnegative portions summing to ℓ only to the attaining physical and corridor inequalities. Do the analogous split for u^+ . At a tie any such split works. At zero width both lower and upper normal directions are available, and the sign choice still satisfies stationarity; no division by the width is used.

All assigned original multipliers satisfy complementarity because each receiving bound is active. They therefore form full fixed-query Lagrange multipliers with the root price and reflected cap multipliers. Keep them fixed while changing only the corridor right-hand sides. A corridor upper bound contributes $\rho^+(u_c + \delta \omega_c)$ and a lower bound contributes $-\rho^-(u_c - \delta \omega_c)$ to the dual objective. Their change gives (43). Weak duality and exact equality at the anchor prove (44).

The certificate's old price labels merely aggregate the original history multipliers. Those multipliers can be assigned to the same public histories under any candidate policy, because the public process is exogenous. Thus the upper bound applies to every new feasible policy, not just policies sharing the old state labels. $\square$

The oracle supplies a supergradient of the fixed-table value in the design parameters. A continuous Benders-style outer master can use these cuts and feasible-query lower bounds, but no finite exact cut count or global positive-release polynomial algorithm follows. The exact graph-sized zero-release formulation, exact fixed-query compiler, and absence of a finite-cut guarantee for the outer design are distinct results.

10 Computational Evidence

10.1 Comparisons, measurement, and audit scopes

All instances are synthetic rational service agreements. The binary comparison family uses discount and payment coefficients one, two public regimes per noninitial date, independent seeded quadratic rewards, and continuation caps above a known feasible constant-tier policy. Thus an exact root promise is available without using any optimizer. The horizon ranges from four to ten dates for the unfolded comparisons, with two seeds at each horizon. A separate public-graph grid comparison extends to twelve dates. Every input and implemented policy is retained.

We implement four explicitly defined methods. The quotient compiler merges coefficient events on the public graph. The independent tree method unfolds the same input, changes variables as in Proposition 2.1, and performs occurrence-indexed exact marginal reduction without public-vertex memoization. It uses a separate interpolation implementation, not the quotient code. The generic numerical comparison applies SciPy’s sparse trust-region interior-point method to the unfolded convex quadratic problem (Virtanen et al., 2020). The primal promise method performs backward dynamic programming directly on the public graph, restricting successor promises to a grid while allowing each current action to be the continuous residual. None of these baselines receives the quotient optimum as an input.

Each case–method pair runs in a fresh sequential worker. Three repetitions are retained with their minimum, median, and maximum. Numerical-library imports are outside timed method calls and are performed in all workers. Peak memory is an absolute process high-water mark, including the common runtime and libraries; it does not estimate allocator-level incremental memory. The comparison suite repeats construction, root inversion, certificate generation, and the inherited whole-response audit separately. In the scale suite, construction, inversion, and machine minimization are repeated three times; certificate generation and an independent single-query policy audit are each timed once and explicitly identified as diagnostics. The latter checks conditional caps, exact state flow, stationarity, complementarity, root payment, and reward without checking every affine response segment. The companion specifies all tolerances, generators, and measurement boundaries.

10.2 What the baselines establish after the R31 reframing

The exact tree comparison is a correctness cross-check, not an algorithmic novelty claim. All eight reported tree optima equal the quotient optima as exact rational numbers. The numerical convex solves terminate successfully; their largest absolute value difference is 3.64×10^{-7} and their largest primal constraint residual is below 2.63×10^{-14} . These residuals are not exact optimality certificates.

For the implemented occurrence-indexed code and these inputs, the tree-to-quotient timing ratio ranges from approximately 1.9 to 40.7. R31 does not use those ratios as evidence against classical laminar algorithms. A normalized memoized occurrence reduction has the same recurrence as the quotient compiler. The computational conclusion from this comparison is exact agreement between two independently implemented representations.

The public-graph promise method is the more informative structural comparator because it already

exploits recombination. Its policies satisfy the exact root promise and every continuation cap under rational replay. At resolution 80, the largest absolute loss among the six reported horizon–seed pairs is below 9.77×10^{-5} ; one instance is already exact at resolution 40. The comparison isolates what the exact dual path contributes beyond graph compression: continuous parametric closure, exact root inversion, and a finite exact execution machine without selecting a promise mesh.

The modern parametric literature discussed in Section 2 is handled theorem by theorem rather than through a mismatched software race. Klimm and Warode (2021), for example, uses a parametric flow input and different parameterization. R31 therefore does not infer practical superiority from a generic QP timing or from an unfolded-tree count. Its computational questions are instead whether the compact compiler reproduces the exact laminar optimum, whether the whole response representation exhibits the proved output-size regimes, and whether one compiled path supports cheap repeated root queries.

10.3 Representation tightness, graph scale, and rational precision

Theorem 3.2 supplies an analytic worst-case family: an n -vertex chain has exactly $2n$ global knots and $n^2 + 2n$ stored affine response segments. This lower bound is independent of implementation and closes the one-sided complexity issue raised by the referee.

The separately retained dense-knot experiment provides an empirical counterpart. With slack continuation caps and distinct local clipping thresholds, a 511-vertex instance stores 197,369 affine segments, compared with 3,161 at 63 vertices. Coefficient lengths remain small and execution needs one writable symbol. The experiment is not used as the proof of quadratic storage; it illustrates the representation regime already established analytically.

A complementary late-renewal family retains heterogeneous binding caps at the penultimate date and recombines afterward. It reaches 2,045 public vertices and 8,164 edges. It has ten global response knots, 22,419 stored segments, and largest coefficient length 2,213 bits. The optimal protocol requires four writable symbols. Construction takes a median 2.62 seconds, while separate certificate generation and single-query audit measurements take 0.87 and 1.19 seconds. This is a long rational-horizon example with sparse cap activation, not a worst-case density test.

The broadly capped coupled family has heterogeneous payment and reward coefficients and rational transitions. Its maximum coefficient length rises from 198 to 388 to 768 bits as the horizon doubles from 16 to 32 to 64 dates; a tighter-cap 128-date instance reaches 1,501 bits. These observations are consistent with, but do not prove, the polynomial bit-height theorem. The theorem itself is algebraic and includes sorting, exact comparisons, and fraction reduction.

The response representation, the reached behavior machine, and the certificate are measured separately. A small writable alphabet does not imply a small read-only response table, and a sparse knot universe does not imply short aggregate certificate numerators. This is why R31 reports K_* , C_* , response segments, coefficient precision, and certificate diagnostics as distinct resources.

10.4 Repeated root queries and the value of compiling the path

A compiled root response supports repeated exact promise inversions without rebuilding all response tables. The retained experiment performs 101 evenly spaced root queries for each late-renewal size. These timings include inversion only; emitting and certifying a new policy for each promise is an additional operation. Corollary 3.5 gives the corresponding exact complexity statement: after preprocessing, root inversion needs logarithmically many exact comparisons in the number of stored knots and a constant number of rational operations.

This repeated-query experiment is aligned with the residual parametric claim. The paper does not say that every application needs the entire path. It establishes when the complete path is compact enough to compile exactly and when that preprocessing can be amortized across root commitments or contractual design queries.

10.5 Verification and the retained evidence boundary

All 35 inherited exact cases reproduce their complete response curves and values, with 147,157 assertion groups in the inherited independent verifier. Forty additional small graph instances agree exactly with the independently implemented tree reduction. Two hundred and two comparisons of explicitly expanded continuation plans check the behavioral equivalence; a separate case reduces two distinct incoming prices to a single necessary symbol. Exhaustive enumeration checks the ordered frontier against 2,308 arbitrary set partitions, including unequal probabilities and nonquadratic costs. Boundary tests cover 150 feasible corridor anchors, 96 tied-normal occurrences, and 1,185 exact supporting inequalities. The test suite also rejects deliberately invalid inputs and altered certificates. These are implementation and finite-instance checks, not independent mathematical proofs of universal claims.

The earlier deployment study is neither removed nor used as favorable algorithmic evidence. At the strict tolerance, 83 of 84 cache queries refreshed and no cache speed advantage was measured. Switching paths, comparator-adjusted rents, continuous-state envelopes, correlated certificates, and governance results retain their original statements, assumptions, proofs, and numerical records in the supplementary theory volumes. The R31 response-complexity theorem and the earlier cache study address different algorithms and different questions.

10.6 New piecewise, circuit, and full-parametric checks

The additional reference implementation uses exact rational arithmetic throughout. Its local optimizer enumerates and compares the maximizing candidate in each quadratic piece; this direct check is separate from the local inverse-marginal response constructor. Twenty-four seeded five-vertex recombining DAGs include differentiable joins, strict marginal jumps, rational caps, and discounting. At all generated event points, intervening prices, and tail probes, 4,170 local optimality comparisons and 834 whole-graph circuit comparisons succeed. The same run verifies 834 circuit root inversions and 836 primal–dual certificates, including fixed-tier and minimum-cap cases. These are finite exact checks of the implementation, not substitutes for the proofs in Section 4.

The parametric reference uses a different mathematical representation: it explicitly unfolds small quadratic problems, enumerates independent active constraint sets, solves the associated rational KKT systems as affine functions of price, and derives the validity interval of each candidate region. It compares coefficients on every open interval of the union of the resulting breakpoints, and also compares boundary values. Each of four three-vertex instances examines 130 candidate active sets; their complete feasible parametric paths have five, three, four, and five regions. All four paths agree exactly with the compact compiler. This is stronger than testing a collection of root promises, but it remains an exhaustive small-instance reference, not a scalable parametric-flow implementation.

The analytic chain is also replayed for sizes two through 128 by powers of two. At the largest size there are 256 global knots and 16,640 explicitly stored response segments. The shared representation instead retains 128 graph nodes and 384 local response segments, together with edges and the event list. The count compares representations of the same function family, not different optima. The reported shared size is persistent post-compilation storage; the test does not claim linear peak preprocessing memory.

Finally, exact partition enumeration returns the deterministic two-symbol loss $1/8$ in Proposition 6.2, and the endpoint lottery returns $1/96$. A rational codebook grid is retained as a regression test only; the global two-symbol optimum follows from the proposition’s piecewise analytic proof. The electronic companion and machine-readable record identify these new checks separately from all inherited experiments.

10.7 Global continuous-codebook and institutional checks

The additional experiments test the global optimization object in Theorem 7.2, not only optimal encoding for a fixed codebook. On 51 exact small instances, the dynamic program agrees with a separately coded exhaustive comparator over all cap-anchored prefixes and analytically minimized terminal intervals. That comparator uses direct branch losses and interpolation of each quadratic, not the production prefix moments. It performs 1,932 candidate comparisons and obtains 446 exact frontier equalities across the randomized and deterministic problems. Another 1,073 direct comparisons use grids containing noncap codewords. Grid agreement is a falsification check; the structural theorem supplies continuous globality.

Independent replay checks 255 expected-participation controllers, including all budgets through eight on four larger inputs. Every branch cap and root promise is verified with zero rational residual. The replay flags individual draws that exceed their branch cap, rather than treating an expectation certificate as a pathwise certificate. Deterministic controllers are separately replayed and realize the pathwise frontier. Table 6 gives the two exact three-branch examples. The heterogeneous example is particularly diagnostic: its optimal final level is not a cap, and a cap-only optimizer returns a strictly larger loss.

The synthetic scale family uses 16–128 branches, positive heterogeneous probabilities, and intermediate costs that include zero curvature. Computing all eight budgets uses the same recurrence; exact output coefficients in this run require at most 29 bits. Timings, complete rational losses, seeds, input conventions, and runtime metadata accompany the source. These tests extend the retained complete-parametric-path checks to a distinct continuous-codebook design problem. They do not report a run of a published specialized parametric solver or establish empirical calibration of service demand.

10.8 Same-input frontier acceleration

The Monge implementation targets the complete restricted-memory frontier on the *same* renewal inputs as the unaccelerated global algorithm. Table 7 reports actual oracle calls and wall times, not a comparison against an exponentially larger unfolding. Both methods use exact fractions and the same primitive moment representation. The new routine changes terminal and prefix search, including leftmost ties and the treatment of missing triangular entries. Every compared budget value agrees exactly.

The companion documents 98 exact instances, 1,712 frontier equalities, 1,712 controller replays, and an independent direct-enumeration comparison. Source-identified negative-input and boundary tests accompany the scale data. The unchanged piecewise-response and R34 global-frontier suites are rerun in temporary directories, leaving their earlier evidence intact. These tests support reproducibility and isolate the structural acceleration; they do not establish empirical service calibration, exhaustive literature priority, or superiority over an unimplemented specialized solver.

10.9 Linear-work search on the same complete frontier

Table 8 compares the implemented linear search with the unchanged monotone search on identical inputs, through 2,048 branches and eight budgets. Both expected and pathwise objectives agree exactly at every size. The new suite adds 100 rational instances, 4,232 old-algorithm objective equalities, 2,116 controller replays, and 43,431 exhaustive submatrix row-minimum checks. A direct branchwise comparator checks 156 reduced-candidate frontier values without production moments or dynamic programming. The companion records all test categories and their limits.

The asymptotic improvement does not imply uniform speed. At 16 branches the new expected-participation search makes 534 economic queries versus 289; at 2,048 it makes 128,966 versus 135,597. At the latter size its deterministic count remains larger: 163,838 versus 153,180. Repeated queries are counted, not hidden behind a cache. Actual runtimes and all exact codebooks are retained; no unexecuted quadratic case is reported as a timeout. This study isolates complete-frontier matrix search, not priority over unimplemented specialized optimization software.

11 Conclusions

Accepted continuations admit an exact compact-graph quotient. Laminar allocation and normalized memoization are classical; the quantitative results concern the response’s event budget, exact rational implementation, and execution architecture. The quadratic specialization has tight linear global-knot and quadratic explicit-table orders. Changing curvature and downward marginal jumps preserve exact closure with a local-piece event budget. Shared circuits reduce persistent read-only space at a query-time cost. None of these storage counts substitutes for additional writable history, characterized by a contract-generated finite transducer and classical behavioral minimization with ordered classes and a linear worst-case alphabet.

Restricted memory is a separate design problem. Deterministic renewal policies admit ordered partitions; expected-participation policies use adjacent-codeword lotteries. The global randomized optimum has cap-

anchored levels except for its continuously optimized highest level. Exact examples show that cap-only discretization misses that optimum and that randomization can reduce restricted-memory loss without changing the zero-loss threshold. Pathwise participation instead restores the deterministic frontier at every budget. These results use the stated renewal architecture and saturated root promise. The radial-spread comparative static concerns its specified ordered family, not a general dispersion order.

Contractual interpolation and interval-minimized terminal costs are Monge despite heterogeneous intermediate curvatures. Their symbolic staircase completion preserves ordered minima in every submatrix. Classical SMAWK then computes either complete frontier with linear work per budget layer, without changing feasible contracts. This improves the preceding quadratic and monotone-search guarantees; measured small-instance overhead and the later expected-frontier crossover remain part of the evidence, rather than a claim of uniform speed superiority.

Exact checks cover piecewise curvature, marginal jumps, shared evaluation, inversion, and conjugate certificates. Independent whole-parametric-path comparisons, reduced-candidate enumeration, and exhaustive submatrix tests target different proof obligations. They do not establish empirical service calibration or superiority over unimplemented specialized methods. Switching, coupled commitments, shared-table supporting cuts, and broader deployment theory retain their separate assumptions and historical evidence.

Code, data, and reproducibility. The repository’s R36 package contains the new proof, linear-search implementations, tests, same-input comparisons, and a point-by-point referee response. The unchanged R33–R35 suites are rerun in temporary directories. All inherited theorem modules and evidence remain available, with the predecessor wrappers archived. The companion documents the algorithm and institutional conventions. A preservation manifest and a full base-tree audit distinguish additions from inherited material; the build binds reader PDFs, test records, and hashes to the exact scientific source commit. Reproduction commands and environment requirements are in the repository README.

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Table 1: Comparison on unfoldable instances

Dates	Seed	Vertices	Occurrences	Quotient	Tree	Convex	Value error
4	13	7	15	0.0009	0.0018	0.0265	3.64e-07
4	29	7	15	0.0009	0.0022	0.0697	9.12e-10
6	13	11	63	0.0023	0.0099	0.0713	5.88e-09
6	29	11	63	0.0022	0.0090	0.1183	8.74e-10
8	13	15	255	0.0040	0.0379	0.2627	1.36e-09
8	29	15	255	0.0039	0.0483	0.3436	2.57e-09
10	13	19	1023	0.0041	0.1663	1.2747	1.39e-08
10	29	19	1023	0.0059	0.1670	2.4538	3.22e-10

Times are seconds. Quotient time is the sum of the separately measured phase medians for construction, root inversion, and certificate generation; the exact audit is excluded. Tree and convex times are end-to-end medians of three repetitions. Tree and quotient values agree as rational numbers. Value error is the absolute difference between the numerical convex-solver value and the exact optimum; its feasibility and stationarity residuals are retained separately. All methods use the same input, not an optimum supplied by another method.

Table 2: Primal promise discretization on the public graph

Dates	Seed	$Q = 10$	$Q = 20$	$Q = 40$	$Q = 80$	Seconds	States
4	13	8.82e-03	5.11e-03	0	0	0.0085	619
4	29	3.21e-02	7.48e-03	1.50e-04	5.29e-06	0.0084	599
8	13	2.54e-02	4.00e-03	3.79e-04	7.69e-05	0.0342	2275
8	29	2.19e-02	8.73e-04	2.25e-04	5.95e-05	0.0333	2279
12	13	1.68e-02	1.87e-03	4.15e-04	9.76e-05	0.0792	4971
12	29	2.15e-02	9.97e-04	3.15e-04	5.68e-05	0.0853	4967

The four resolution columns report the exact feasible-policy value loss relative to the quotient optimum. Child promises are multiples of $1/Q$; local actions are continuous residuals and the root promise is not rounded. Seconds and stored states refer to $Q = 80$. Times are medians of three repetitions. The grid is a public-graph dynamic program, not a scenario-tree baseline; exact rational replay checks its feasibility. A zero gap on a particular instance is not a uniform exactness claim for the grid.

Table 3: Response representation and rational coefficient growth

Family	Vertices	Edges	Knots	Segments	Largest	Bits
Coupled	61	228	128	2428	83	198
Coupled	125	484	240	10168	159	388
Coupled	253	996	436	34748	279	768
Tight caps	509	2020	723	67212	295	1501
Late renewal	253	996	10	2706	11	277
Late renewal	509	2020	10	5523	11	554
Late renewal	1021	4068	10	11154	11	1107
Late renewal	2045	8164	10	22419	11	2213
Dense knots	63	122	96	3161	68	12
Dense knots	127	250	192	12473	132	14
Dense knots	255	506	384	49529	260	16
Dense knots	511	1018	768	197369	516	18

Knots counts the distinct global union; segments counts the entire stored family. Bits is the largest numerator or denominator bit length among reduced affine coefficients. Late-renewal caps are heterogeneous and active only at the penultimate date; this family is deliberately distinct from the dense-knot stress family and from the broadly capped coupled family. It tests long rational horizons with a small knot universe, not worst-case quadratic storage.

Table 4: Separate construction, execution, and audit costs

Family	Vertices	Build	Invert	Certificate	Audit	Symbols	Memory
Coupled	61	0.1220	16.3	0.0484	0.0775	18	97.3
Coupled	125	0.5158	183.1	0.1154	0.2222	23	117.5
Coupled	253	2.3536	433.9	0.6680	1.1607	61	227.2
Tight caps	509	7.1123	1426.9	0.5700	0.1233	1	396.5
Late renewal	253	0.0976	19.1	0.0154	0.0293	4	97.2
Late renewal	509	0.2551	24.8	0.0483	0.0839	4	107.4
Late renewal	1021	0.7465	48.9	0.1691	0.2958	4	137.3
Late renewal	2045	2.6156	127.1	0.8716	1.1871	4	251.0
Dense knots	63	0.0417	69.7	0.0043	0.0037	1	94.3
Dense knots	127	0.1679	146.2	0.0163	0.0092	1	100.2
Dense knots	255	0.6180	271.4	0.1215	0.0153	1	121.4
Dense knots	511	2.5074	504.5	0.3282	0.0315	1	214.3

Build, certificate, and audit are seconds; inversion is microseconds. Build and inversion use three repetitions. Certificate generation and the independent single-query policy audit are each measured once in this scale suite; they are diagnostic measurements, not repeated-time claims. Symbols is the exact minimized writable alphabet. Memory is the absolute fresh-worker process high-water mark in MiB, including common Python, NumPy, and SciPy imports; it is not incremental algorithm allocation. Whole-curve audits are reported separately from these single-query KKT/flow audits.

Table 5: New exact verification scopes. Counts refer to one complete run.

Object	Count	Verified property
Piecewise recombining inputs	24	Valid local joins and caps
Local direct optimization checks	4,170	Exact maximizing action
Common-price circuit comparisons	834	All vertex payments
Circuit root inversions	834	Exact promised payment
Primal–dual certificates	836	Feasibility and equality
Complete active-set paths	4	Every affine region
Chain representation sizes	7	Exact knot and piece counts
Two-symbol renewal instance	1	Deterministic and lottery losses

Table 6: Exact two-symbol frontiers under two participation institutions

Branch probabilities	Expected loss	Pathwise loss	Optimal expected levels
(1/3, 1/3, 1/3)	1/96	1/8	(1/4, 3/4)
(7/20, 3/5, 1/20)	23/1280	3/160	(1/4, 5/8)

Both rows use caps $(1/4, 1/2, 3/4)$, terminal reward $2c - c^2/2$, and intermediate cost $y^2/2$. Pathwise losses equal the deterministic optimum by Theorem 7.4. Three symbols attain zero loss in both institutions. In the second row, restricting randomized levels to caps gives 3/160, not the unrestricted 23/1280. The first row is the inherited exact example; the second is the new continuous-level regression.

Table 7: Exact complete-frontier comparison on identical renewal inputs. All budgets one through eight are computed. Counts are tail-interval and prefix-edge oracle calls, not bit operations.

Branches	R34 calls	Accelerated calls	R34 seconds	Accelerated seconds
16	546	289	0.015	0.010
32	2,594	842	0.064	0.023
64	11,298	2,177	0.266	0.057
128	47,138	5,247	1.068	0.130
256	192,546	12,177	4.515	0.310
512	—	27,600	—	0.718
1,024	—	61,546	—	1.642

The first five rows have exact equality with the unchanged R34 complete-frontier algorithm. Dashes mark unexecuted baseline runs, not timeouts or estimated results. Larger rows pass exact controller replay. All instances are synthetic.

Table 8: Complete expected-participation frontiers on identical inputs. All budgets one through eight are computed. Counts include repeated economic oracle evaluations, but not ordinal padding comparisons.

Branches	Monotone calls	Linear calls	Monotone seconds	Linear seconds
16	289	534	0.008	0.014
32	842	1,447	0.022	0.033
64	2,177	3,375	0.054	0.073
128	5,247	7,342	0.127	0.156
256	12,177	15,376	0.299	0.341
512	27,600	31,551	0.697	0.721
1,024	61,546	63,985	1.584	1.493
2,048	135,597	128,966	3.559	3.129

Monotone denotes the preceding divide-and-conquer implementation; linear denotes the implemented SMAWK search with symbolic staircase completion. Every row has exact equality for both expected and pathwise frontiers and direct controller replay. The quadratic randomized baseline agrees through 256 branches; larger quadratic runs are not executed. Timing is machine-dependent. Inputs are synthetic.